



UNIT II

Roots of Equation

Syllabus :

Motivation, Bracketing methods: Bisection methods, Open methods : Newton Raphson method, Engineering applications.

Syllabus Topic : Motivation

2.1 Introduction

- In Applied mathematics and Engineering, various problems are in the form of determining the root of an equation.
- A great importance is given to such problems which are in the form

$$f(x) = 0$$

Where $f(x)$ is the function, either linear or non-linear.

$$\text{Let, } f(x) = ax^2 + bx + c = 0.$$

$$\text{and } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- The value of x which we obtain from the above equation, when substituted in the $f(x)$ function, then we get value of the function $f(x) = 0$. So the value of x is called the root of the function.

$$\text{e.g. Let } y = f(x) = 4x - 8.$$

$$\text{Now when } x = 2, f(x) = 0. \text{ i.e. } y = 0.$$

$\therefore x$ is the root of the function $(4x - 8)$.

2.1.1 Graphical Representation

- Let the function be, $y = f(x) = x^2 + 2x - 1$.
- Then root of the function lie where $f(x) = 0$ or $y = 0$.

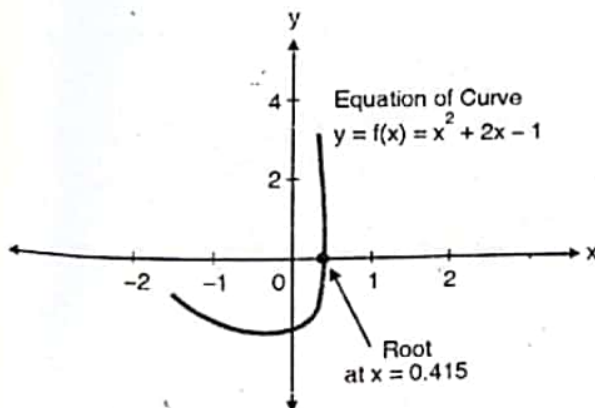


Fig. 2.1.1

- From the Fig. 2.1.1 it is clear that the point at which curve $y = f(x)$ intersect x axis is the root of the equation. i.e. at $x = 0.415$ because for every point on the x axis, the value of y is zero.
- As the curve $y = f(x) = x^2 + x + 1$ intersect x axis at $x = 0.415$.
- Therefore the root of the equation is **0.415**.

2.2 Methods to Obtain the Root of Equation

- The solution to find the root of the equation $ax^2 + bx + c = 0$ by analytical method is,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Where x is root of the equation

- There are also trial and error or graphical methods available to find the root of the equation.
- But with the Iterative method and with the use of computer we get root of the equation very easily.
- The Iterative method is continued till the given number of iterations or the desired value of accuracy.
- Iterative methods are classified as :

Classification of Iterative methods

- (1) Bracketing method or Interpolation Method
- (2) Open end method or Extrapolation method

Fig. C2.1 : Classification of iterative methods

2.2.1 Bracketing Method or Interpolation Method

- In this method, we assume two initial guesses in such a way that the root lies between these two guesses. Therefore we restrict the root, or bracket the root in between two guesses.
- e.g. $f(x) = x^2 - 8x + 16 = 0$. The root of the equation is 4 (Analytically) Now we assume two initial guesses.



i. e. two values of x in such a way that, the root lies between these two values.

i. e. Let $x_1 = 1^{\text{st}}$ Initial guess = 3

and $x_2 = 2^{\text{nd}}$ Initial guess = 5

- Therefore we restrict the value of root in between 3 and 5.
- The selection of proper initial guesses and then the approach towards the root can be done by bracketing methods. i.e Bisection or Half interval method.

2.2.2 Open End Method or Extrapolation Method

- In this method, we assume a single initial guess which is closer to the root of the equation. But we don't restrict the root, or bracket the root.
- Method to find the root of the equation using single guess is Newton - Raphson method
- In case of Bracketing method, we have assumed two initial guesses such that bracket the root and then we reduced the width of the bracket until we reached closer to the root.
- In case of Open method, we use a single starting guess or two guesses that do not necessarily bracket the root.
- Open method, we sometimes move away from the actual root or we diverge from the true root while calculating the answer.

Syllabus Topic : Bracketing Methods - Bi-section Method

2.3 Bi-section or Half Interval Method

- This method is also called binary chopping or Bolzano's method.
- This is the most easiest iterative method for finding the root of the equation.
- If two initial guesses are given as x_1 and x_2 , then the first approximate value of root of the equation lie at the mid value of x_1 and x_2 .
- Let x_3 be the first approximate value of the root, then according to the method as the name suggest, bisection or half interval, $x_3 = \frac{x_1 + x_2}{2}$.

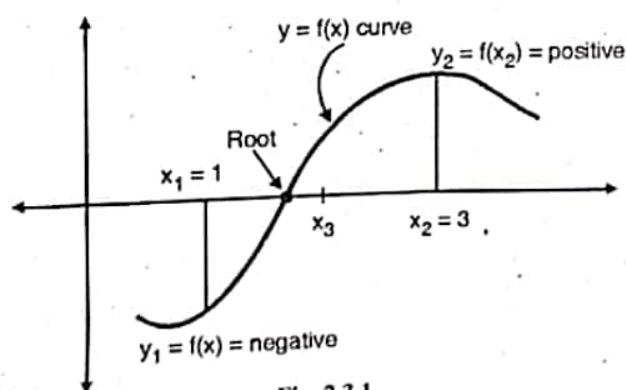


Fig. 2.3.1

Procedure

Let the curve be $y = f(x)$

The actual root lies where the curve intersect the X-axis,

i.e. at $y = f(x) = 0$

- (i) We have to assume initial guess x_1 and x_2 in such a way that $f(x_1)$ i. e. y_1 and $f(x_2)$ i. e. y_2 both have alternate sign.

(i. e. one positive and one negative)

or we can say $f(x_1) \times f(x_2) < 0$

or $y_1 \times y_2 < 0$

e.g. If $y = f(x) = x^2 + 8x - 4$ and initial guesses are $x_1 = 0$ and $x_2 = 1$ then,

The value of $f(x_1)$ can be obtained by substituting the value of $x_1 = 0$ in the $f(x)$ equation..

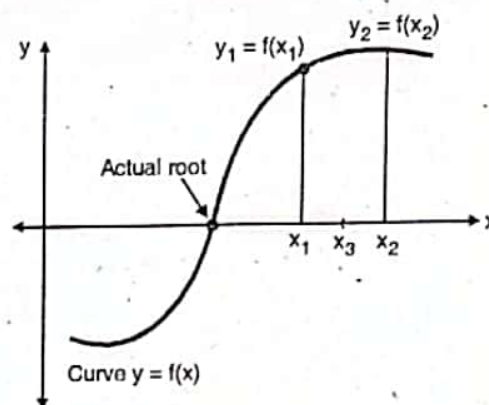


Fig. 2.3.2

$$\therefore f(x_1) = x_1^2 + 8x_1 - 4 = 0^2 + 8(0) - 4 = -4 \text{ (negative)}$$

$$\therefore y_1 = f(x_1) = \text{negative}$$

The value of $f(x_2)$ can be obtained by substituting the value of $x_2 = 1$ in the $f(x)$ equation.

$$\therefore f(x_2) = x_2^2 + 8x_2 - 4 = 1^2 + 8(1) - 4 = 5 \text{ (positive)}$$

$$\therefore y_2 = f(x_2) = \text{positive.}$$

If $y_1 \times y_2 < 0$ then only our assumed guesses x_1 and x_2 are right.

- As shown in Fig. 2.3.2. If we consider, the initial guesses x_1 and x_2 in such a way that $y_1 = f(x_1)$ and $y_2 = f(x_2)$ both are positive i.e. $f(x_1) \times f(x_2) > 0$.

- Then the root $x_3 = \frac{x_1 + x_2}{2}$ lies between the interval of x_1 and x_2 which is no where closer to the root.

- i.e. when root of the function $f(x)$ will not lie in between x_1 and x_2 , we will never get the actual root.



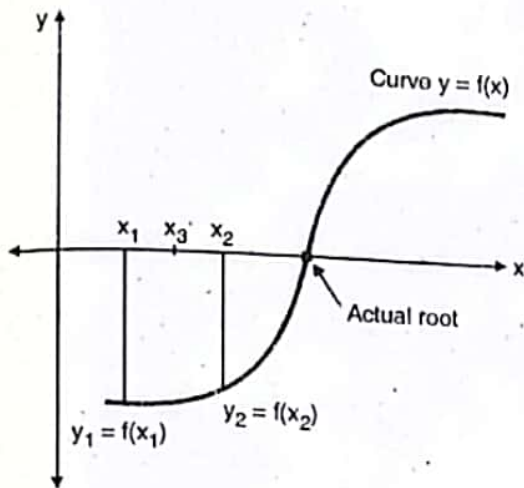


Fig. 2.3.3

- As shown in Fig. 2.3.3. Now, if we consider the initial guesses x_1 and x_2 in such a way that $y_1 = f(x_1)$ and $y_2 = f(x_2)$ are both negative. i.e. $f(x_1) \times f(x_2) > 0$.
- Then the root $x_3 = \frac{x_1 + x_2}{2}$ lies between the interval of x_1 and x_2 which is again nowhere closer to the root i.e. when root of the function $f(x)$ will not lie in between x_1 and x_2 , will never get the actual root.

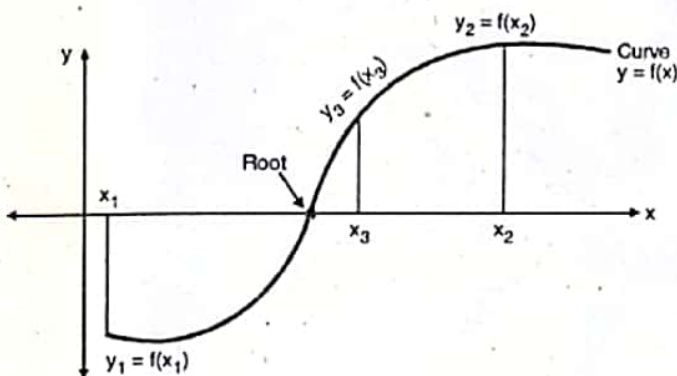


Fig. 2.3.4

- From the above two cases, we can say that, the right approach towards the root can be possible if our initial guesses x_1 and x_2 are right or in other words $f(x_1)$ and $f(x_2)$ should have opposite sign i.e. $f(x_1) \times f(x_2) < 0$ as shown in Fig. 2.3.4.
- (ii) When the condition $f(x_1) \times f(x_2) < 0$ is satisfied, then, we can approach towards the actual root using bisection method. i.e. 1st approximate value of root $x_3 = \frac{x_1 + x_2}{2}$ as shown in Fig. 2.3.4.
- (iii) Now we find value of $y_3 = f(x_3)$ by substituting the value of x_3 in $y = f(x)$ equation.
- (iv) From the Fig. 2.3.4 it is cleared that value of $y_3 = f(x_3)$ is positive. Now we have to replace previous guess x_1 or x_2 with the first approximate root x_3 in such a way that either $f(x_1) \times f(x_3) < 0$ or $f(x_2) \times f(x_3) < 0$

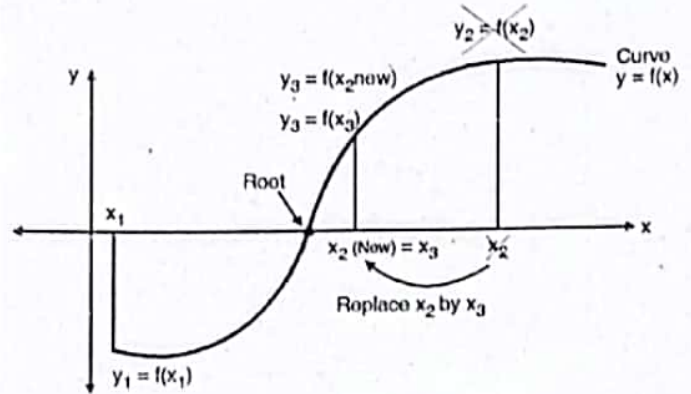


Fig. 2.3.5

- (v) From Fig. 2.3.5, as $f(x_3)$ is positive and $f(x_2)$ is also positive, hence $f(x_1) \times f(x_3) < 0$. So the previous assume guess x_2 is replaced by x_3 .

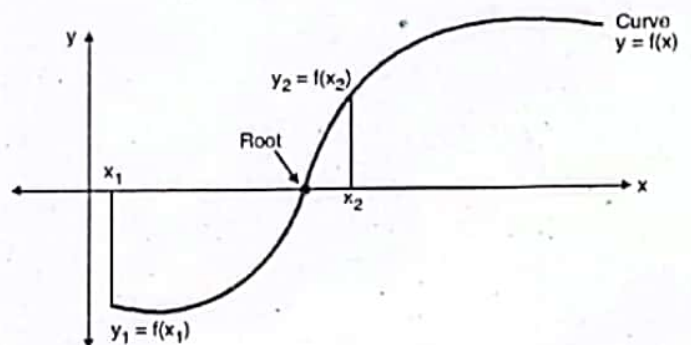


Fig. 2.3.6

- (vi) From Fig. 2.3.6, the new initial guesses for calculation of 2nd approximate new root x_3 will be $x_1 = x_1$ and $x_2 = x_3$

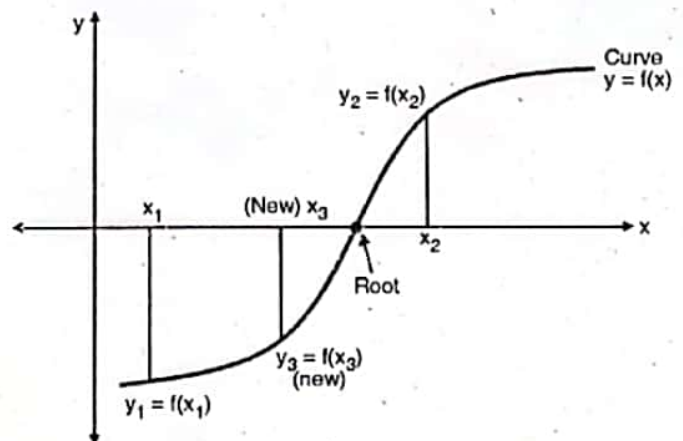


Fig. 2.3.7

- (vii) From Fig. 2.3.7, according to bi section method, the second approximate value of root, $x_3 = \frac{x_1 + x_2}{2}$ Again we find value of $y_3 = f(x_3)$ by substituting value of x_3 in $y = f(x)$ equation.

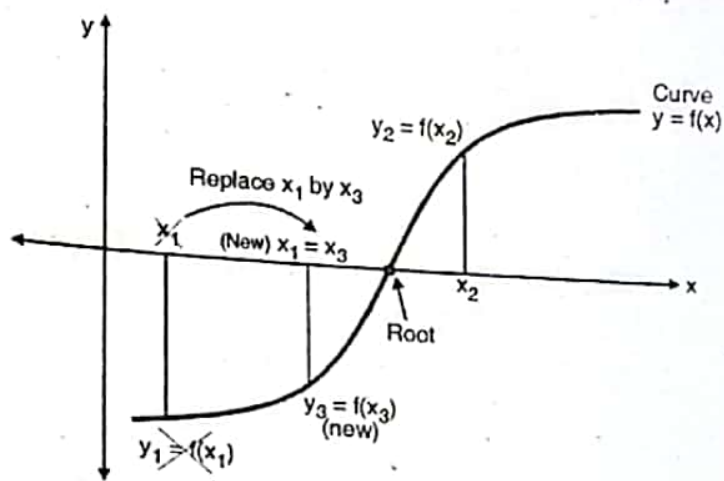


Fig. 2.3.8

(viii) From the Fig. 2.3.8 it is clear that new value of $y_3 = f(x_3)$ is negative. Now we have to replace previous guesses x_1 and x_2 with the new value x_3 such that either,

$$f(x_1) \times f(x_3) < 0 \quad \text{or} \quad f(x_2) \times f(x_3) < 0$$

As $f(x_3)$ is negative, hence $f(x_2) \times f(x_3) < 0$.

So the previous guess x_1 is replaced by new value of x_3 .

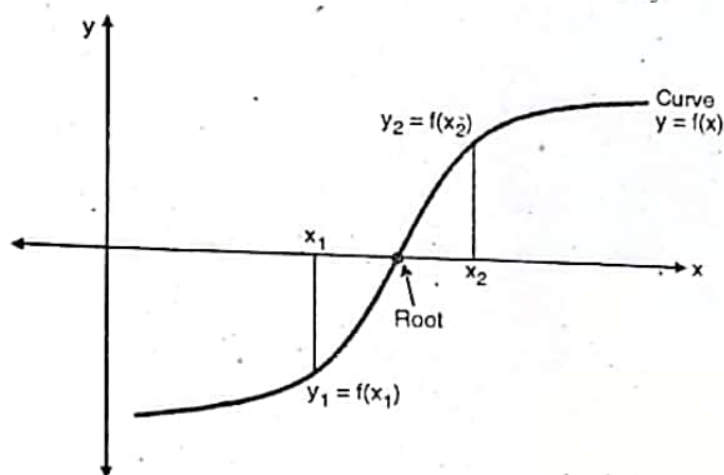


Fig. 2.3.9

(ix) From Fig. 2.3.9 the initial guesses for the calculation of 3rd approximate new root x_3 will be $x_1 = x_3$ and $x_2 = x_2$.

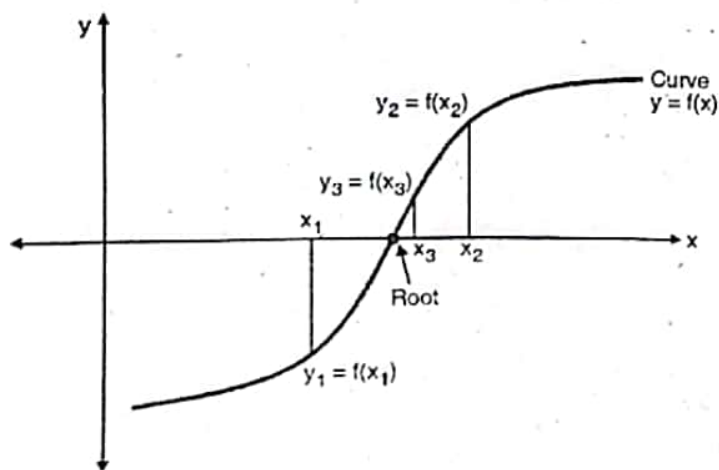


Fig. 2.3.10

(x) From Fig. 2.3.10, according to bi-section method the third approximate value of root $x_3 = \frac{x_1 + x_2}{2}$. Again we find value of $y_3 = f(x_3)$ by substituting value x_3 in $y = f(x)$ equation.

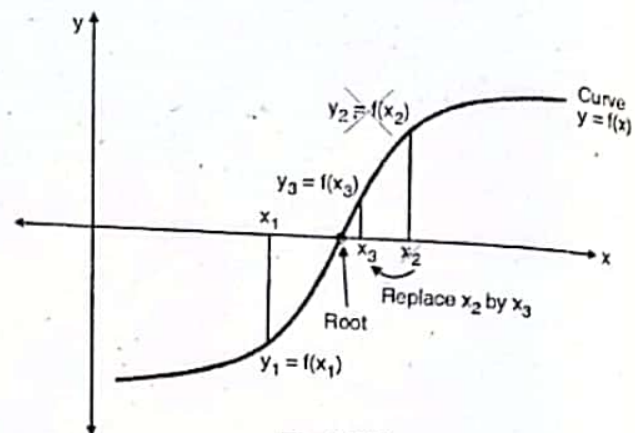


Fig. 2.3.11

(xi) From the Fig. 2.3.11 it is clear that new value of $y_3 = f(x_3)$ is positive. Now we have to replace previous guesses x_1 and x_2 with the new value x_3 such that either,

$$f(x_1) \times f(x_3) < 0 \quad \text{or} \quad f(x_2) \times f(x_3) < 0$$

As $f(x_3)$ is positive, hence $f(x_1) \times f(x_3) < 0$.

So the previous guess x_2 is replaced by new value of x_3 .

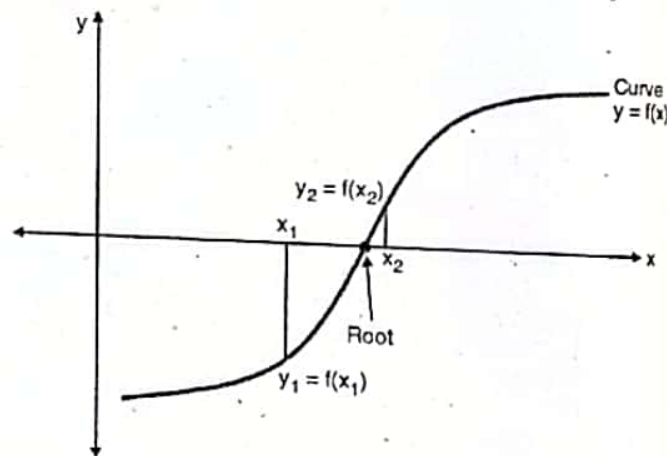


Fig. 2.3.12

(xii) From Fig. 2.3.12 the initial guesses for the calculation of 3rd approximate new root x_3 will be $x_2 = x_3$ and $x_1 = x_1$.

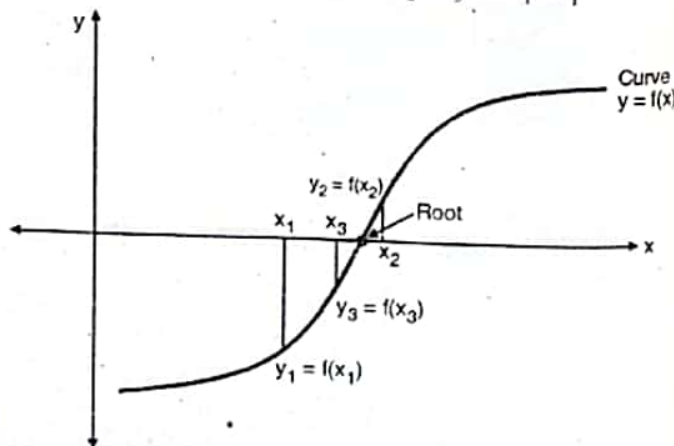


Fig. 2.3.13



(xiii) From Fig. 2.3.13, according to bi-section method the third approximate value of root $x_3 = \frac{x_1 + x_2}{2}$. Again we find value of y_3 , again we check the sign of $f(x_1)$ and $f(x_2)$ again we replace value of x_3 with either x_1 or x_2 .

The above procedure is repeated till the number of iterations or till the desired accuracy is achieved $|x_2 - x_1| < \text{Accuracy}$.

2.3.1 Stepwise Solution to Solve the Problems on Bisection Method(Accuracy Criteria) (Illustrated by Numerical Example)

Example 2.3.1

Determine using bi-section method, a root of the equation $\cos x - 1.3x = 0$ with accuracy of 0.01.

Solution

Given data $f(x) = \cos x - 1.3x$. No initial guesses are given.

So assume, Initial guess as, $x_1 = 0$

and $x_2 = 1$

Accuracy = 0.01

Step 1: Check whether assume initial guesses are right or wrong.

For $x_1 = 0$

$$\begin{aligned} f(x_1) &= \cos x_1 - 1.3x_1 \\ &= \cos(0) - 1.3(0) = 1 = \text{positive} \end{aligned}$$

For $x_2 = 1$

$$\begin{aligned} \therefore f(x_2) &= \cos(x_2) - 1.3(x_2) \\ &= \cos(1) - 1.3(1) \\ &= 0.54 - 1.3 = -0.76 = \text{negative} \end{aligned}$$

Now, $f(x_1) \times f(x_2) < 0$ (i.e. both have opposite signs)

Hence the initial guesses x_1 and x_2 are right

$\therefore x_1 = 0, x_2 = 1$ and $f(x_1) = \text{positive}$,

$f(x_2) = \text{negative}$

ITERATION 1

Step 2: Apply bi-section method's iterative formula.

If x_3 is the new root

$$\therefore x_3 = \frac{x_1 + x_2}{2} = \frac{0 + 1}{2} = 0.5$$

Step 3: Check whether $f(x_3)$ is either positive or negative.

$$\begin{aligned} \therefore f(x_3) &= \cos(x_3) - 1.3(x_3) \\ &= \cos(0.5) - 1.3(0.5) = 0.2275 \end{aligned}$$

$\therefore f(x_3) = \text{positive}$

Step 4: Replace value of x_1 or x_2 by x_3 (whose $f(x)$ i.e. $f(x_1)$ or $f(x_2)$ will have the same sign as that of $f(x_3)$).

This is because we have to satisfy the condition either $f(x_1) \times f(x_3) < 0$ or $f(x_2) \times f(x_3) < 0$.

As $f(x_1) = \text{positive}$,

$f(x_2) = \text{negative and}$

$f(x_1) = \text{positive}$

$\therefore$ Replace x_1 guess with the value of x_3 .

So that $f(x_2) \times f(x_3) < 0$

or $f(x_2) \times f(x_1) < 0$

$\therefore x_1 = x_3 = 0.5$ and $y_1 = y_3 = 0.22759$ positive

$\therefore$ The new guesses are $x_1 = 0.5$ and $x_2 = 1$.

With $f(x_1) = \text{positive and } (x_2) = \text{negative}$

Step 5: Check whether $|x_2 - x_1| < \text{Accuracy}$ or not.

$\therefore |0.5 - 1| < \text{Accuracy}$.

or $0.5 > 0.01$

$\therefore$ Repeat step 2 and step 5 till we get $|x_2 - x_1| < \text{Accuracy}$.

ITERATION 2

Step 2: Apply bi-section method's iterative formula.

$$x_3 = \frac{x_1 + x_2}{2} = \frac{0.5 + 1}{2} = 0.75$$

Note: Convert the calculator on 'RAD' mode when the problem contains trigonometric function like (cos, sin)

Step 3: Check $f(x_3)$ is either positive or negative.

$$\begin{aligned} \therefore f(x_3) &= \cos(0.75) - 1.3(0.75) \\ &= -0.2433 \end{aligned}$$

$f(x_3) = \text{negative}$.

Step 4: Replace x_1 or x_2 by x_3 (whose $f(x)$ have the same sign as that of $f(x_3)$)

$f(x_1) = \text{positive}$,

$f(x_2) = \text{negative and } f(x_3) = \text{negative}$

As $f(x_2)$ have negative sign $\therefore x_2$ is replaced by x_3

$\therefore x_2 = x_3 = 0.75$

and $y_2 = y_3 = -0.24$ (negative)

New guesses are $x_1 = 0.5$ and $x_2 = 0.75$

With $f(x_1) = \text{positive and } f(x_2) = \text{negative}$

Step 5: Check whether $|x_2 - x_1| < \text{Accuracy}$ or not.

$|x_2 - x_1| = 0.75 - 0.5 = 0.25$

$\therefore |x_2 - x_1| > \text{Accuracy } 0.25 > 0.01$

Repeat step 2 and step 5 till $|x_2 - x_1| < \text{Accuracy}$.

ITERATION 3

Step 2: Apply bi-section method's iterative formula.

$$x_3 = \frac{x_1 + x_2}{2} = \frac{0.5 + 0.75}{2} = 0.625$$

Step 3: Check $f(x_3)$ is either positive or negative.

$$\begin{aligned} f(x_3) &= \cos(x_3) - 1.3(x_3) \\ &= \cos(0.625) - 1.3(0.625) \\ &= -0.001537 \end{aligned}$$

$\therefore f(x_3) = \text{negative}$.

Step 4: Replace x_1 or x_2 by x_3 (whose $f(x)$ have the same sign as that of $f(x_3)$)



$f(x_1)$ = positive, $f(x_2)$ = negative,

$f(x_3)$ = negative

As $f(x_2)$ have the same sign as that of $f(x_3)$.

$\therefore$ Replace x_2 by x_3 .

$$\therefore x_2 = x_3 = 0.625$$

and $y_2 = y_3 = -0.0015$ (negative)

$\therefore$ New guesses are $x_1 = 0.5$ and $x_2 = 0.625$

with $f(x_1)$ = positive and $f(x_2)$ = negative.

Step 5: Check whether $|x_2 - x_1| < \text{Accuracy}$ or not.

$$\therefore 0.625 - 0.5 > \text{Accuracy}$$

$$\therefore 0.125 > 0.01$$

Repeat step 2 and step 5 till $|x_2 - x_1| < \text{Accuracy}$.

ITERATION 4

Step 2: Apply bi-section method's iterative formula.

$$x_3 = \frac{x_1 + x_2}{2} = \frac{0.5 + 0.625}{2} = 0.5625.$$

Step 3: Check $f(x_3)$ is either positive or negative.

$$\begin{aligned} f(x_3) &= \cos(x_3) - 1.3(x_3) \\ &= \cos(0.5625) - 1.3(0.5625) \\ &= 0.114675. \end{aligned}$$

$\therefore f(x_3)$ = positive.

Step 4: Replace x_1 or x_2 by x_3 (whose $f(x)$ have the same sign as that of $f(x_3)$)

$f(x_1)$ = positive, $f(x_2)$ = negative.

$f(x_3)$ = positive.

As $f(x_1)$ have the same sign as that of $f(x_3)$.

$\therefore$ Replace x_1 by x_3

$\therefore x_1 = x_3 = 0.5625$ and $y_1 = y_3 = 0.1146$ (positive)

New guesses are $x_1 = 0.5625$ and $x_2 = 0.625$

with $f(x_1)$ = positive and $f(x_2)$ = negative.

Step 5: Check whether $|x_2 - x_1| < \text{Accuracy}$ or not

$$\therefore 0.625 - 0.5625 > \text{Accuracy}$$

$$\text{or } 0.0625 > \text{Accuracy}$$

Repeat the above steps till $|x_2 - x_1| < \text{Accuracy}$

ITERATION 5

Step 2: Apply bi-section method's iterative formula.

$$x_3 = \frac{x_1 + x_2}{2} = \frac{0.5625 + 0.625}{2} = 0.59375.$$

Step 3: Check $f(x_3)$ is either positive or negative.

$$\begin{aligned} f(x_3) &= \cos(x_3) - 1.3(x_3) \\ &= \cos(0.59375) - 1.3(0.59375) \\ &= 0.056973 \end{aligned}$$

$\therefore f(x_3)$ = positive.

Step 4: Replace x_1 or x_2 by x_3 (whose $f(x)$ have the same sign as that of $f(x_3)$)

$f(x_1)$ = positive, $f(x_2)$ = negative, $f(x_3)$ = positive.

$\therefore$ Replace x_1 by x_3 as $f(x_3)$ = positive.

$$\therefore x_1 = x_3 = 0.59375$$

and $y_1 = y_3 = 0.0569$ (positive)

New guesses are $x_1 = 0.59375$ and $x_2 = 0.625$

With $f(x_1)$ = positive and $f(x_2)$ = negative.

Step 5: Check $|x_2 - x_1| < \text{Accuracy}$ or not

$$\therefore 0.625 - 0.59375 > \text{Accuracy}$$

$$\text{or } 0.031 > \text{Accuracy}$$

Repeat the above steps 2 to 5 till $|x_2 - x_1| < \text{Accuracy}$.

ITERATION 6

Step 2: Apply bi-section method's iterative formula.

$$\begin{aligned} x_3 &= \frac{x_1 + x_2}{2} = \frac{0.59375 + 0.625}{2} \\ &= 0.609375. \end{aligned}$$

Step 3: Check $f(x_3)$ is either positive or negative.

$$\begin{aligned} f(x_3) &= \cos(x_3) - 1.3(x_3) = 0.0278 \\ &= \cos(0.6093) - 1.3(0.6093) \\ &= 0.02781 \end{aligned}$$

$\therefore f(x_3)$ = positive.

Step 4: Replace x_1 or x_2 by x_3 (whose $f(x)$ have the same sign as that of $f(x_3)$)

$f(x_1)$ = positive,

$f(x_2)$ = negative, $f(x_3)$ = positive

$\therefore$ Replace x_1 by x_3

$$\therefore x_1 = x_3 = 0.609375$$

and $y_1 = y_3 = 0.02781$ (positive)

$\therefore$ New guesses are $x_1 = 0.609375$ and $x_2 = 0.625$

with $f(x_1)$ = positive and $f(x_2)$ = negative.

Step 5: Check $|x_2 - x_1| < \text{Accuracy}$ or not.

$$\therefore 0.625 - 0.609375 > \text{Accuracy}$$

$$\text{or } 0.0156 > 0.01$$

Repeat the above steps till $|x_2 - x_1| < \text{Accuracy}$

ITERATION 7

Step 2: Apply bi-section method's iterative formula.

$$\begin{aligned} x_3 &= \frac{x_1 + x_2}{2} \\ &= \frac{0.609375 + 0.625}{2} = 0.617188 \end{aligned}$$

Step 3: Check $f(x_3)$ is either positive or negative.

$$\begin{aligned} f(x_3) &= \cos(x_3) - 1.3(x_3) = 0.0131 \\ &= 0.0131 = \text{positive} \end{aligned}$$

$\therefore f(x_3)$ = positive

Step 4: Replace x_1 or x_2 by x_3 (whose $f(x)$ have the same sign as that of $f(x_3)$)

$f(x_1)$ = positive,

$f(x_2)$ = negative, $f(x_3)$

= positive

Replace x_1 by x_3 as $f(x_3)$ = positive

$x_1 = x_3 = 0.617188$
 and $y_1 = y_3 = 0.0131$ (positive)
 $\therefore$ New guesses are $x_1 = 0.617188$ and $x_2 = 0.625$
 with $f(x_1) = \text{positive}$
 and $f(x_2) = \text{negative}$

Step 5: Check $|x_2 - x_1| < \text{Accuracy}$ or not

$$\therefore 0.625 - 0.617188 = 0.0078$$

$$\therefore 0.0078 < 0.01$$

$\therefore$ The final guesses are $x_1 = 0.617188$ and $x_2 = 0.625$.

ITERATION 8

Step 2: Apply bi-section method's iterative formula.

$\therefore$ Root of the equation

$$x_3 = \frac{x_1 + x_2}{2} = \frac{0.617188 + 0.625}{2}$$

$$x_3 = 0.621094$$

The result is tabulated below

Iteration number	x_1	$y_1 = f(x_1)$	x_2	$y_2 = f(x_2)$	x_3	$y_3 = f(x_3)$	Accuracy
1	0	1	1	-0.76	0.5	0.2275	0.5
2	0.5	0.2275	1	-0.76	0.75	-0.2433	0.25
3	0.5	0.2275	0.75	-0.2433	0.625	-0.001537	0.125
4	0.5	0.2275	0.625	-0.001537	0.5625	0.1146	0.0625
5	0.5625	0.1146	0.625	-0.001537	0.5937	0.05697	0.031
6	0.5937	0.05697	0.625	-0.001537	0.6093	-0.027	0.0156
7	0.6093	0.027	0.625	-0.001537	0.6171	0.0131	0.0078
8	0.6171	0.0131	0.625	-0.001537	0.6210		↑

Accuracy

$\therefore$ The root of the equation for given accuracy is

$$x_3 = 0.6210$$

...Ans.

Note: Flowchart and Program on Bisection method (Accuracy criteria) are explained in Chapter 9.

Example 2.3.2

Solve the equation, $xe^x - 3.1(\cos x) = 0$. by bisection method up to an accuracy of 0.01

Solution:

Given data: $f(x) = x \cdot e^x - 3.1(\cos x)$

$$\text{Accuracy} = 0.1$$

No Initial guesses are given

So we assume $x_1 = 0$, $x_2 = 1$

Step 1: Check whether initial guesses are right or wrong.

$$\text{For } x_1 = 0 \quad \therefore f(x_1) = 0 \cdot e^0 - 3.1(\cos(0))$$

$$= -2.1 = \text{negative}$$

$$\text{For } x_2 = 1 \quad \therefore f(x_2) = 1 \cdot e^1 - 3.1(\cos(1)) = 1.043$$

$$= \text{positive}$$

$\therefore$ As $f(x_1) \times f(x_2) < 0$ (i.e. both have opposite signs)

our assumed guesses are right.

ITERATION 1

Step 2: Apply bi-section method to calculate x_3

$$x_3 = \frac{x_1 + x_2}{2} = \frac{0 + 1}{2} = 0.5$$

Step 3: Check $f(x_3)$ is positive or negative.

$$\therefore f(x_3) = (0.5) \cdot e^{(0.5)} - 3.1 \cdot \cos(0.5)$$

$$= -1.896 = \text{negative}$$

Step 4: Replace x_1 or x_2 by x_3

$$\therefore f(x_1) = \text{negative}, f(x_2) = \text{positive},$$

$$f(x_3) = \text{negative}$$

As $f(x_1)$ have negative sign

$\therefore x_1$ is replace by x_3

$$\therefore x_1 = x_3 = 0.5 \quad \text{and}$$

$$y_1 = y_3 = -1.896 \text{ (negative)}$$

Step 5: Check $|x_2 - x_1| < \text{Accuracy}$ or not.

$$\therefore |1 - 0.5| > \text{Accuracy}$$

$$\text{or } 0.5 > 0.01.$$

Repeat Step 2 to 4 till we get $|x_2 - x_1| < \text{Accuracy}$

ITERATION 2

Step 2: Apply bi-section method to calculate x_3

$$x_3 = \frac{x_1 + x_2}{2} = \frac{0.5 + 1}{2} = 0.75$$

Step 3: Check whether $f(x_3)$ is either positive or negative.

$$\therefore f(x_3) = x_3 \cdot e^{x_3} - 3.1(\cos x_3)$$

$$= (0.75) e^{(0.75)} - 3.1 \cos(0.75)$$

$$= -0.680485 = \text{negative}$$

Step 4: Replace x_1 or x_2 by x_3

$$\therefore f(x_1) = \text{negative}, f(x_2) = \text{positive}, f(x_3) = \text{negative}$$

As $f(x_1)$ have negative sign

$\therefore x_1$ is replace by x_3

$$\therefore x_1 = x_3 = 0.75$$

$$\therefore y_1 = y_3 = -0.680485$$

Therefore new guesses are $x_1 = 0.75$ and $x_2 = 1$

Step 5: Check whether $|x_2 - x_1| < \text{Accuracy}$ or not

$$\therefore |0.75 - 1| > \text{Accuracy}$$

$$0.25 > 0.01$$

$\therefore$ Repeat the steps from 2 to 4 till $|x_2 - x_1| < \text{Accuracy}$



The remaining iterations are tabulated below

Iteration number	x_1	$y_1 = f(x_1)$	x_2	$y_2 = f(x_2)$	x_3	$y_3 = f(x_3)$	Accuracy
1	0	-2.100000	1	1.043345	0.5	-1.896145	0.5
2	0.5	-1.896145	1	1.043345	0.75	-0.680485	0.25
3	0.75	-0.680485	1	1.043345	0.875	0.111926	0.1255
4	0.75	-0.680485	0.875	0.111926	0.81250	-0.300828	0.0625
5	0.8125	-0.300828	0.875	0.111926	0.84375	-0.098686	0.031
6	0.84375	-0.098686	0.875	0.111926	0.859	0.005548	0.015
7	0.84375	-0.098686	0.859	0.005548	0.851	-0.0468	0.008
8	0.851	-0.0468	0.859	0.005548	0.8554	← Root	↑ Accuracy

∴ The root of the equation for given accuracy is

$$x_3 = 0.8554$$

...Ans.

2.3.2 Stepwise Solution to Solve the Problems on Bisection Method (Iteration Criteria) (Illustrated by Numerical Example)

Example 2.3.3

Using three iteration of bi-section method determine root of the equation. Initial guesses are :

$$x_1 = 2.8 \text{ and } x_2 = 3, f(x) = -0.9x^2 + 1.7x + 2.5.$$

Solution :

$$\text{Given data : } f(x) = -0.9x^2 + 1.7x + 2.5 \quad \dots(i)$$

Initial guess $x_1 = 2.8$ and $x_2 = 3$

Number of iteration, $n = 3$

Step 1 : Check whether initial guesses are right or wrong.

For $x_1 = 2.8$

$$\begin{aligned} \therefore f(x_1) &= -(0.9) \times (2.8)^2 + 1.7(2.8) + 2.5 \\ &= 0.204 = \text{positive} \end{aligned}$$

(∵ From Equation (i))

$$\begin{aligned} \text{For } x_2 = 3 \quad f(x_2) &= -0.9(3)^2 + 1.7(3) + 2.5 \\ &= -0.5 = \text{negative} \end{aligned}$$

(∵ From Equation (i))

$$\therefore f(x_1) \times f(x_2) < 0 \quad (\text{i.e. both have opposite signs})$$

∴ Our initial guesses are right.

ITERATION 1

Step 2 : Apply bisection method formula

$$\therefore \text{New root, } x_3 = \frac{x_1 + x_2}{2} = \frac{2.8 + 3}{2} = 2.9$$

Step 3 : Check $f(x_3)$ is either positive or negative.

$$\begin{aligned} \therefore f(x_3) &= -0.9(2.9)^2 + 1.7(2.9) + 2.5 \\ &= -0.139 = \text{negative} \end{aligned}$$

Step 4 : Replace x_1 or x_2 by x_3

$$\begin{aligned} f(x_1) &= \text{positive, } f(x_2) = \text{negative} \\ f(x_3) &= \text{negative} \end{aligned}$$

As $f(x_2)$ have negative sing

∴ x_2 is replaced by x_3

$$\therefore x_2 = x_3 = 2.9$$

$$\text{and } y_2 = y_3 = -0.139 \text{ (negative)}$$

Repeat step 2 to 4 till number of iteration.

The problem is based on iteration criteria. Then skip step-5

ITERATION 2

Step 2 : Apply bisection method formula

$$x_3 = \frac{x_1 + x_2}{2} = \frac{2.8 + 2.9}{2}$$

$$x_3 = 2.85$$

Step 3 : Check $f(x_3)$ either positive or negative.

$$\begin{aligned} \therefore f(x_3) &= -0.9(2.85)^2 + 1.7(2.85) + 2.5 \\ &= 0.0347 = \text{positive} \end{aligned}$$

Step 4 : Replace x_1 or x_2 by x_3

$$f(x_1) = \text{positive, } f(x_2) = \text{negative}$$

$$f(x_3) = \text{positive}$$

As $f(x_1)$ have positive sign

∴ x_1 is replace by x_3

$$\therefore x_1 = x_3 = 2.85$$

$$\text{and } y_1 = y_3 = 0.0347 \text{ (positive).}$$

ITERATION 3

Step 2 : Apply bisection method formula

$$x_3 = \frac{x_1 + x_2}{2} = \frac{2.85 + 2.9}{2} = 2.875$$

Step 3 : Check $f(x_3)$ is positive or negative

$$\begin{aligned} \therefore f(x_3) &= -0.9(2.875)^2 + 1.7(2.875) + 2.5 \\ &= -0.051562 = \text{negative} \end{aligned}$$

Step 4 : After 3 iteration the guesses are

$$x_1 = 2.85, \quad x_2 = 2.9 \quad \text{and} \quad x_3 = 2.875$$

The result is tabulated as below

Iteration number	x_1	$y_1 = f(x_1)$	x_2	$y_2 = f(x_2)$	x_3	$y_3 = f(x_3)$
1	2.8	0.204	3	-0.5	2.9	-0.139
2	2.8	0.204	2.9	-0.139	2.85	0.0345
3	2.85	0.0345	2.9	-0.139	2.875	-
						0.051562

∴ The root after 3rd iteration

$$x_3 = 2.875.$$

...Ans.

Note : Flowchart and Program on Bisection method (Iteration criteria) are explained in Chapter 9.

Example 2.3.4

Using five iteration of bi-section method determine root of the equation $f(x) = x^3 - x^2 - x - 1 = 0$. Assume initial guess $x_1 = 1.7$ and $x_2 = 1.9$.

Solution :

Given data : $f(x) = x^3 - x^2 - x - 1$

Initial guess $x_1 = 1.7$ and $x_2 = 1.9$

Number of iteration, $n = 5$

Step 1 : Check whether initial guesses are right or wrong.

$$\text{For } x_1 = 1.7 \quad \therefore f(x_1) = x_1^3 - x_1^2 - x_1 - 1$$

$$= -0.677 = \text{negative}$$

$$\text{For } x_2 = 1.9 \quad f(x_2) = x_2^3 - x_2^2 - x_2 - 1 = 0.349$$

$$= \text{positive}$$

$$\therefore f(x_1) \times f(x_2) < 0 \text{ (i.e. both have opposite signs)}$$

∴ Our Initial guesses are right.

ITERATION 1

Step 2 : Apply bisection method formula

$$\therefore \text{New root, } x_3 = \frac{x_1 + x_2}{2} = \frac{1.7 + 1.9}{2} = 1.8$$

Step 3 : Check $f(x_3)$ is either positive or negative.

$$\therefore f(x_3) = x_3^3 - x_3^2 - x_3 - 1 = -0.208$$

$$= \text{negative}$$

Step 4 : Replace x_1 or x_2 by x_3

$$f(x_1) = \text{negative}, f(x_2) = \text{positive}, f(x_3) = \text{negative}$$

$$f(x_1), f(x_3) \text{ have same sign.} \quad \therefore x_1 \text{ is replace by } x_3$$

$$\therefore x_1 = x_3 = 1.8$$

$$\text{and } y_1 = y_3 = -0.208 \text{ (negative)}$$

Repeat step 2 to 4 till number of iteration.

The remaining iterations are tabulated as below

Iteration number	x_1	$y_1 = f(x_1)$	x_2	$y_2 = f(x_2)$	x_3	$y_3 = f(x_3)$
1	1.7	-0.677	1.9	0.349	1.8	-0.208
2	1.8	-0.208	1.9	0.349	1.85	0.0591
3	1.8	-0.208	1.85	0.0591	1.825	-0.07723
4	1.825	-0.07723	1.85	0.0591	1.84375	-0.009759
5	1.825	-0.07723	1.84375	-0.009759	1.8437	0.024

∴ The root after 5th iteration $x_3 = 1.8437$.

...Ans.

Syllabus Topic : Open Method - Newton Raphson Method

2.4 Open Method

- In case of Open method, we use a single starting guess or two guesses that do not necessarily bracket the root.
- Open method, we sometimes move away from the actual root or we diverge from the true root while calculating the answer.
- Open Method is Newton Raphson method

2.4.1 Newton Raphson Method

- In this type of open method, we assume or use a single starting guess to find the solution.
- The value of new guess or root is estimated by drawing a tangent at point y_1 whose co-ordinates are (x_1, y_1) . The tangent intersect x-axis at a particular point which represents the new value of guess x_2 .

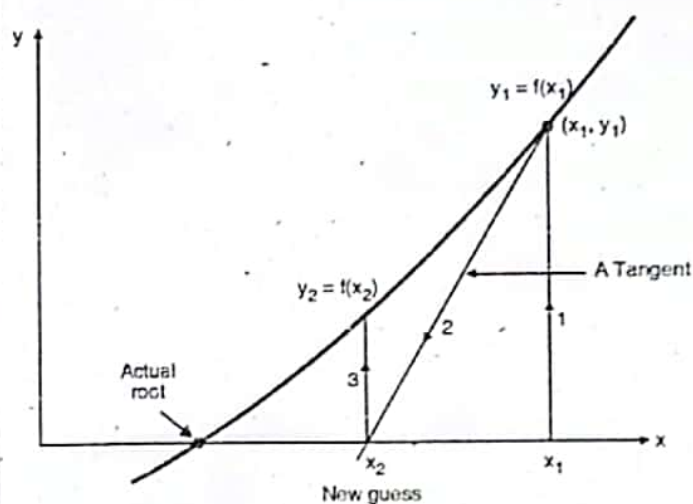


Fig. 2.4.1

- As shown in Fig. 2.4.1, from the initial guess x_1 , we find the value of $y_1 = f(x_1)$. A tangent from point y_1 of co-ordinate (x_1, y_1) is drawn which cut the x-axis at point x_2 . ∴ x_2 is the new guess for further calculation.
- The equation of tangent by tangent point method is,

$$\frac{y - y_1}{x - x_1} = m = f'(x_1)$$
- The tangent intersect x-axis at point $(x_2, 0)$

- ∴ Hence point $(x_2, 0)$ satisfy the equation of tangent
 ∴ The equation becomes,

$$\begin{aligned}\therefore \frac{0 - y_1}{x_2 - x_1} &= f'(x_1) \\ \therefore -y_1 &= f'(x_1) \cdot (x_2 - x_1) \\ \therefore -y_1 &= f'(x_1) \times x_2 - f'(x_1) \cdot x_1 \\ \therefore x_2 &= \frac{-y_1 + f'(x_1) \cdot x_1}{f'(x_1)} \\ \therefore x_2 &= x_1 - \frac{y_1}{f'(x_1)} \\ \therefore \text{but } y_1 &= f(x_1) \\ \therefore x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ \text{or } x_2 &= x_1 - \frac{y_1}{f'(x_1)}\end{aligned}$$

Procedure

Let the curve be $y = f(x)$.

- i) We have to assume initial guess x_1 in such a way that,

$$\frac{f(x_1) \times f''(x_1)}{[f'(x_1)]^2} \leq 1, \text{ to have the convergence.}$$

- The details about convergence is explained later in the section.
- If we want to approach or converge towards the root, the above condition should be satisfied.
- ii) When our assume guess x_1 is right, then we can approach towards the actual root by Newton-Raphson method.

∴ The new value of guess $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ is then calculated.

- Then the new value of guess becomes initial guess for the next calculation.

$$\therefore x_1 = x_2 \text{ and } y_1 = f(x_1)$$

Again we calculate value of new guess

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

As shown in Fig. 2.4.2,

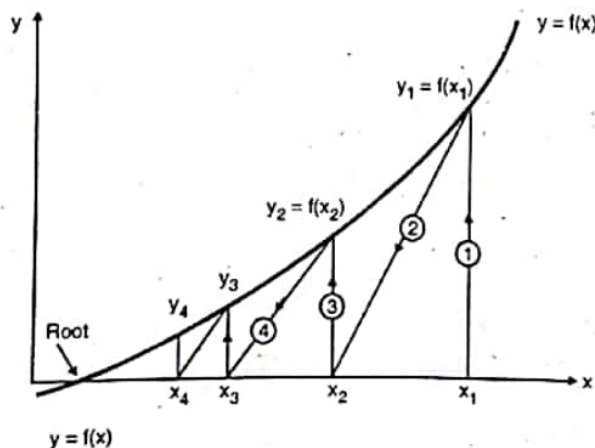


Fig. 2.4.2

Path 1: It indicates from assume guess x_1 , the value of y_1 is calculated as $y_1 = f(x_1)$.

Path 2: It indicates the new value of x_2 which can be calculated by equation, (or by drawing tangent at y_1)

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

The value of x_2 becomes initial guess for next iteration,

$$\therefore x_1 = x_2 \text{ and } y_1 = y_2$$

Path 3: It indicates from value of x_2 we calculate again value of $y_2 = f(x_2)$

$$\text{or } y_1 = f(x_1) \quad (\because x_1 = x_2)$$

Path 4: It indicates the new value of x_2 which can be calculated by equation.

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

- By approaching in the above manner we are going closer to the actual root as shown in Fig. 2.4.2.
- We repeat the procedure till the number of iteration or till the desired accuracy is achieved.

$$\text{i.e. } |x_2 - x_1| < \text{Accuracy}$$

$$\text{or } |x_{2\text{new}} - x_{2\text{old}}| < \text{Accuracy}$$

2.4.2 Convergence and Divergence In Case Of Newton - Raphson Method

Review Questions

- Write a short note on concept of convergence.
- Explain the convergence and divergence in Newton Raphson method.
- What do you mean by convergence? Explain importance in brief?
- Explain the concept of convergence in Newton Raphson method.

Convergence

- As discussed earlier in Newton-Raphson method, the root is estimated by drawing a tangent at point y_1 whose co-ordinate are (x_1, y_1) .
- The tangent intersect x-axis at a point which represent the root of the curve $y = f(x)$.
- The approach towards the root is shown in Fig. 2.4.3.

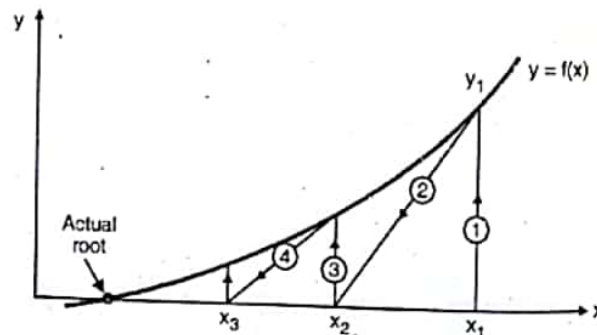


Fig. 2.4.3

- As shown in Fig. 2.4.3 the path 1 to 4 shows that, with Newton-Raphson method, we are approaching towards the actual root.
- Hence the method is converges or we can say convergence occurs in the above case.

The value of the root x_2 can be calculated as,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = g(x_1)$$

The slope of is less than 1. To have the convergence $|g'(x_1)| < 1$

$$\therefore g(x_1) = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Differentiating above equation w.r.t. x_1

$$\therefore g'(x_1) = 1 - \left[\frac{f'(x_1) \cdot f'(x_1) - f(x_1) \cdot f''(x_1)}{[f'(x_1)]^2} \right]$$

$$\therefore g'(x_1) = 1 - \left[\frac{[f'(x_1)]^2 - f(x_1) \cdot f''(x_1)}{[f'(x_1)]^2} \right]$$

$$\therefore g'(x_1) = 1 - \left[\frac{1 - f(x_1) \cdot f''(x_1)}{[f'(x_1)]^2} \right]$$

$$= \frac{f(x_1) \cdot f''(x_1)}{[f'(x_1)]^2}$$

$\therefore$ for the convergence, $|g'(x)| < 1$

or for the convergence, $\left| \frac{f(x_1) \cdot f''(x_1)}{[f'(x_1)]^2} \right| < 1$

Divergence

As shown in Fig. 2.4.4, approach towards the root become difficult by Newton-Raphson method.

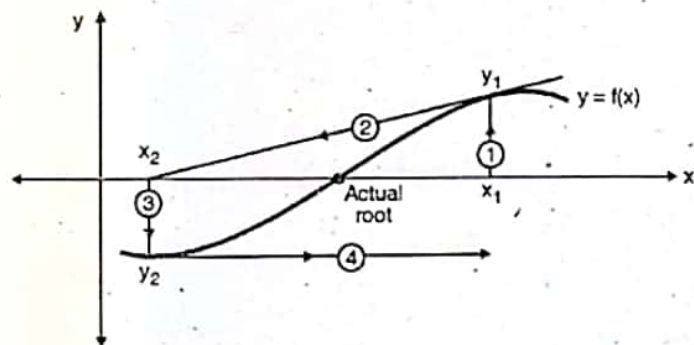


Fig. 2.4.4 (a)

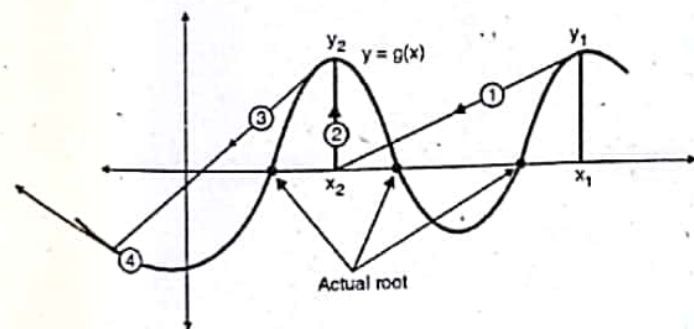


Fig. 2.4.4 (b)

- From Fig. 2.4.4 (a) and (b) it is cleared that by drawing tangent, we are no where moving closer to the root.

- We are actually moving away from the root as the slope of curve is greater than 1.
- Hence the method is diverges or we can say divergence is occur in the above case.

$\therefore$ In both the cases

$$|g'(x_1)| > 1$$

Condition for the divergence $|g'(x_1)| > 1$

or Condition for the divergence $\left| \frac{f(x_1) \cdot f''(x_1)}{[f'(x_1)]^2} \right| > 1$

2.4.3 Stepwise Solution to Solve the Problems on Newton Raphson Method(Accuracy Criteria) (Illustrated by Numerical Example)

Example 2.4.1

Solve using Newton-Raphson Method, the equation $e^x \cdot \cos x - 1.4 = 0$. Find the value of the root upto the accuracy of 0.001.

Solution :

Given data : $f(x) = e^x \cdot \cos x - 1.4$

Accuracy = 0.001

Let the Initial guess $x_1 = 0$.

Step 1 : Calculate value of $f'(x)$:

$$\therefore f(x) = e^x \cos x - 1.4 \quad \dots(i)$$

Differentiating above equation w. r. to x .

$$\begin{aligned} \therefore f'(x) &= [e^x \cos x + e^x (-\sin x)] - 0 \\ &= e^x \cos x - e^x \sin x \\ f'(x) &= e^x (\cos x - \sin x) \quad \dots(ii) \end{aligned}$$

Step 2 : Calculate value of $f''(x)$:

Differentiating equation (ii) again w. r. to x .

$$\begin{aligned} \therefore f''(x) &= e^x (\cos x - \sin x) + e^x (-\sin x - \cos x) \\ &= e^x [\cos x - \sin x - \sin x - \cos x] \\ &= e^x [-2 \sin x] \\ f''(x) &= -2 e^x \sin x \quad \dots(iii) \end{aligned}$$

Step 3 : Check whether assume initial guess is right or wrong :

$\therefore$ According to the condition for convergence,

$$\frac{f(x_1) \times f''(x_1)}{[f'(x_1)]^2} \leq 1$$

$\therefore$ From Equation (i) $f(x_1) = f(0) = e^0 \cdot \cos(0) - 1.4$

$$f(x_1) = -0.4$$

from Equation (ii) $f'(x_1) = f'(0) = e^0 (\cos(0) - \sin(0))$

$$f'(x_1) = 1$$

from Equation (iii) $f''(x_1) = f''(0) = e^0 [-2 \sin(0)]$

$$f''(x_1) = 0$$

$$\therefore \frac{f(x_1) \times f''(x_1)}{[f'(x_1)]^2} = \frac{-0.4 \times 0}{1}$$

$$\therefore 0 < 1$$

Our assume Initial guess $x_1 = 0$ is right.

Note: When the Initial guess is given in the problem, then we do not have to check the Initial guess whether it is right or wrong.

Then skip Step - 2 and 3 and go for the Step - 4.

ITERATION 1

Step 4: Calculate the new value of root x_2 using Newton-Raphson Method:

$$\begin{aligned}\therefore x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 0 - \left(\frac{-0.4}{1} \right) = 0.4\end{aligned}$$

Step 5: Check whether $|x_2 - x_1| < \text{Accuracy}$ or not:

$$\begin{aligned}\therefore |0.4 - 0| &> \text{Accuracy} \\ \text{or } 0.4 &> 0.001\end{aligned}$$

Step 6: If the accuracy is not achieved then replace, x_1 by x_2

$\therefore$ New Initial guess will be $x_1 = x_2$

or $x_1 = 0.4$

Repeat Step 4 to 6 till we get $|x_2 - x_1| < \text{Accuracy}$.

ITERATION 2

Step 4: Calculate the new value of root x_2 using Newton-Raphson Method

$$\begin{aligned}\text{The new root } x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ \text{where } f(x_1) &= e^{0.4} \cdot \cos(0.4) - 1.4 = -0.0259 \\ \text{and } f'(x_1) &= e^{0.4} (\cos(0.4) - \sin(0.4)) \\ &= 0.7931\end{aligned}$$

$$\begin{aligned}\therefore x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 0.4 - \frac{(-0.0259)}{0.7931} = 0.4327\end{aligned}$$

Step 5: Check whether $|x_2 - x_1| < \text{Accuracy}$ or not.

$$\begin{aligned}\therefore |0.4327 - 0.4| &> \text{Accuracy} \\ \text{or } 0.032 &> 0.001\end{aligned}$$

Step 6: New initial guess will be $x_1 = x_2$

or $x_1 = 0.4327$

ITERATION 3

Step 4: Calculate the new value of root x_2 using Newton-Raphson Method

$$\begin{aligned}\text{The new root } x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ \text{where, } f(x_1) &= e^{0.4327} (\cos(0.4327) - 1.4) \\ &= -6.47 \times 10^{-4} \\ \text{and } f'(x_1) &= e^{0.4327} (\cos(0.4327) - \sin(0.4327)) \\ &= 0.753\end{aligned}$$

$$\begin{aligned}\therefore x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 0.4327 - \left(\frac{-6.47 \times 10^{-4}}{0.753} \right) = 0.4336\end{aligned}$$

Step 5: Check whether $|x_2 - x_1| < \text{Accuracy}$ or not:

$$\begin{aligned}\therefore |0.4336 - 0.4327| &< \text{Accuracy} \\ \text{or } 0.0009 &< 0.001 \\ \therefore x_2 &= 0.4336\end{aligned}$$

The result is tabulated below

Iteration Number	x_1	$f(x_1)$	$f'(x_1)$	x_2	Accuracy
1	0	-0.4	1	0.4	0.4
2	0.4	-0.0259	0.7931	0.4327	0.032
3	0.4327	-6.47×10^{-4}	0.753	0.4336	0.0009

$\therefore$ The root of equation upto the accuracy of 0.001 is $x_2 = 0.4336$
...Ans

Note: Flowchart and Program on Newton Raphson Method (Accuracy criteria) are explained in Chapter 9.

Example 2.4.2

Find the root of the equation $\sin x - x \cos x$. Using Newton-Raphson Method Assume Initial guess as $x_1 = 3\pi/2$ upto accuracy of 0.00001.

Solution:

Given data: $f(x) = \sin x - x \cdot \cos x$
Initial guess = $3\pi/2 = 4.7124$
Accuracy = 0.00001

Step 1: Calculate value of $f'(x)$.

$$\therefore f(x) = \sin x - x \cdot \cos x$$

Differentiating above equation w. r. t. x .

$$\begin{aligned}\therefore f'(x) &= \cos x - [-x \cdot \sin x + \cos x] \\ &= \cos x + x \sin x - \cos x\end{aligned}$$

$$\therefore f'(x) = x \sin x$$

Skip Step 2 and 3 as the initial guess $x_1 = 3\pi/2$ is given.

ITERATION 1

Step 4: The new value of root using Newton-Raphson Method.

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\text{Now, } f(x_1) = \sin(4.712) - (4.712) \cdot \cos(4.712) = -1$$

$$\text{and } f'(x_1) = (4.712) \cdot \sin(4.712) = -4.712$$

$$\therefore x_2 = 4.712 - \left(\frac{-1}{-4.712} \right) = 4.500$$

Step 5: Check whether $|x_2 - x_1| < \text{Accuracy}$ or not:

$$\begin{aligned}\therefore |4.5 - 4.712| &> \text{Accuracy} \\ 0.212 &> 0.00001\end{aligned}$$

Step 6: Replace x_1 by x_2 :

$$\therefore x_1 = x_2 = 4.5$$

New Initial guess $x_1 = 4.5$

Repeat Step 4 to 6 till we get $|x_2 - x_1| < \text{Accuracy}$.
The remaining iterations are tabulated as below.

Iteration number	x_1	$y_1 = f(x_1)$	$f'(x_1)$	x_2	Accuracy
2	4.5	-0.0297	-4.399	4.49342	0.0067
3	4.493	-0.000045	-4.386125	4.49341	0.00001

The root for the desired accuracy is $x_2 = 4.4934$...Ans.

Example 2.4.3

Find the root of equation $x^3 - \cos^2(x) = 0$, using Newton-Raphson method, upto accuracy 0.02. Explain Convergence Criteria for this method.

Solution:

Given Data: $f(x) = x^3 - \cos^2(x) = 0$

Accuracy = 0.02

let initial guess $x_1 = 1$

Step 1: Calculate value of $f'(x)$

$$f(x) = x^3 - \cos^2(x) \quad \dots(i)$$

Differentiating above equation w.r.t.x.

$$\begin{aligned} f'(x) &= 3x^2 - 2\cos x (-\sin x) \\ &= 3x^2 + 2\cos x \sin x \quad \dots(ii) \end{aligned}$$

Step 2: Calculate value of $f''(x)$

Differentiating Equation (ii) again w.r.t.x.

$$f''(x) = 6x + 2(\cos x \cdot \cos x - \sin x \sin x) \quad \dots(iii)$$

Step 3: Check whether assume initial guess is right or wrong.

$\therefore$ According to the condition for convergence,

$$\frac{f(x_1) \times f''(x_1)}{[f'(x_1)]^2} \leq 1$$

$$\therefore \text{From Equation (i)} \quad y_1 = f(x_1) = f(1) = 0.70807$$

$$\text{From Equation (ii)} \quad y' = f'(x_1) = 3.9092$$

$$\text{From Equation (iii)} \quad y'' = f''(x_1) = 5.1677$$

$$\therefore a = \frac{f(x_1) \times f''(x_1)}{[f'(x_1)]^2} = 0.2394 < 1$$

Our assume initial guess $x_1 = 1$ is right.

Step 4: Calculate the new value of root x_2 using Newton-Raphson method:

$$\begin{aligned} \therefore x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 1 - \left(\frac{0.70807}{3.9092} \right) = 0.8188 \end{aligned}$$

Step 5: Check whether $|x_2 - x_1| < \text{Accuracy}$ or not

$$\therefore |0.8188 - 1| > \text{Accuracy}$$

Step 6: If accuracy is not achieved then replace x_1 by x_2 .

$\therefore$ New initial guess will be $x_1 = x_2$.

$$\text{or } x_1 = 0.8188$$

Repeat step 4 to 6 till we get $|x_2 - x_1| < \text{Accuracy}$.

Remaining iterations are tabulated as below.

Iteration Number	x_1	$f(x_1)$	$f'(x_1)$	x_2	Accuracy
2	0.8188	0.08232	3.0090	0.791443	0.02735
3	0.791443	0.00179	2.8790	0.790821	0.000622

$\therefore$ The root upto accuracy of 0.02 is 0.790821. ...Ans.

Example 2.4.4

Use Newton's method to find the smallest root of the equation $e^x - \sin(x) = 0$ to four decimal places of accuracy.

Solution:

Given Data: $f(x) = e^x - \sin(x)$

Accuracy = 0.00001

let initial guess $x_1 = -3$

Step 1: Calculate value of $f'(x)$

$$f(x) = e^x - \sin(x) \quad \dots(i)$$

Differentiating above equation w.r.t.x.

$$f'(x) = e^x - \cos(x) \quad \dots(ii)$$

Step 2: Calculate value of $f''(x)$

Differentiating Equation (ii) again w.r.t.x.

$$f''(x) = e^x + \sin(x) \quad \dots(iii)$$

Step 3: Check whether assume initial guess is right or wrong.

$\therefore$ According to the condition for convergence,

$$\frac{f(x_1) \times f''(x_1)}{[f'(x_1)]^2} \leq 1$$

$$\therefore \text{From Equation (i)} \quad y_1 = f(x_1) = f(-3) = 0.1909$$

$$\text{From Equation (ii)} \quad y' = f'(x_1) = 1.0397$$

$$\text{From Equation (iii)} \quad y'' = f''(x_1) = -0.09133$$

$$\therefore a = \frac{f(x_1) \times f''(x_1)}{[f'(x_1)]^2} = -0.016 < 1$$

Our assume initial guess $x_1 = -3$ is right.

Step 4: Calculate the new value of root x_2 using Newton-Raphson method:

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = -3 - \left(\frac{0.1909}{1.0397} \right) = -3.1836$$

Step 5: Check whether $|x_2 - x_1| < \text{Accuracy}$ or not

$$\therefore |-3.1836 - (-3)| > \text{Accuracy}$$

Step 6: If accuracy is not achieved then replace x_1 by x_2 .

$\therefore$ New initial guess will be $x_1 = x_2$.

$$\text{or } x_1 = -3.1836$$

Repeat step 4 to 6 till we get $|x_2 - x_1| < \text{Accuracy}$.

Remaining iterations are tabulated as below.

Iteration No.	x_1	$f(x_1)$	$f'(x_1)$	x_2	Accuracy
2	-3.1836	0.08255	-0.00056	-3.18306	0.00054
3	-3.18306	0	1.04059	-3.183063	0.000007

$\therefore$ The root upto accuracy of 0.00001 is -3.183063 ...Ans.

Example 2.4.5

Apply Newton Raphson method to determine the roots of the equation

$$f(x) = \cos x - x e^x = 0 \text{ to an accuracy of } 0.0001.$$

Solution :

Given data : $f(x) = \cos x - x \cdot e^x$

Accuracy = 0.0001

Let Initial guess $x_1 = 1$.

Step 1 : Calculate value of $f'(x)$:

As, $f(x) = \cos x - x e^x$

$$\therefore f(x_1) = \cos(1) - 1 \times e^1 = -2.1779$$

$$\therefore f'(x) = -\sin x - [x e^x + e^x]$$

$$\therefore f'(x_1) = -\sin(1) - [1 \times e^1 + e^1] \quad (\because x_1 = 1)$$

$$= -6.278$$

Step 2 : Calculate value of $f''(x)$:

$$\therefore f'(x) = -\cos x - [x e^x + e^x]$$

$$= -\cos x - [x e^x + 2 e^x]$$

$$f'(x_1) = -\cos 1 - [1 \times e^1 + 2 e^1]$$

$$= -8.6951$$

Step 3 : Check whether assume initial guess is right or wrong :

$$\therefore \frac{f(x_1) \times f''(x_1)}{[f'(x_1)]^2} \leq 1$$

$$\therefore \frac{-2.1780 \times (-8.6951)}{(-6.278)^2} = 0.4805$$

$\therefore$ As $0.4805 < 1$, our assume initial guess $x_1 = 1$ is right.

ITERATION 1

Step 4 : Calculate the new value of root x_2 using Newton-Raphson method :

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1 - \frac{(-2.1780)}{(-6.278)}$$

$$= 0.653079$$

Step 5 : Check whether $|x_2 - x_1| < \text{Accuracy}$ or not :

$$\therefore |0.653079 - 1| > \text{Accuracy}$$

$$\text{or } 0.347 > 0.0001$$

Step 6 : If accuracy is not achieved then replace x_1 by x_2 :

$$\therefore \text{New initial guess will be } x_1 = x_2$$

$$\text{or } x_1 = 0.653079$$

Repeat steps 4 to 6 till we get $|x_2 - x_1| < \text{Accuracy}$

The remaining iterations are tabulated as follow :

Iteration No.	x_1	$f(x_1)$	$f'(x_1)$	x_2	Accuracy
02	0.653079	-0.4604	-3.7839	0.531343	-0.1217
03	0.531343	-0.0418	-3.1118	0.51791	-0.01343
04	0.51791	-4.6×10^{-4}	-3.0429	0.517757	0.000153
05	0.517757	-5.9×10^{-8}	-3.0421	0.51775	0.00002

$\therefore$ the root upto the accuracy of 0.0001 is $x_2 = 0.51775$

Example 2.4.6

Find a real root of the equation, $e^x \cos(x) - 1.2 = 0$ by the Newton Raphson method correct to three decimal places. State how do you select initial value of x .

Solution :

Given data :

$$y = f(x) = e^x \cos x - 1.2$$

initial guess = $x_1 = 0$

Step 1 : Calculate value of $f'(x)$

$$f(x) = e^x \cos x - 1.2 \quad \dots(1)$$

Differentiating w.r.t. 'x'

$$\therefore f'(x) = [e^x \cos x + e^x (-\sin x)] - 0 = e^x \cos x - e^x \sin x$$

$$\therefore f'(x) = e^x (\cos x - \sin x) \quad \dots(2)$$

Step 2 : Calculate value of $f''(x)$

Differentiating Equation (2) again w.r.t 'x'

$$\therefore f''(x) = e^x (\cos x - \sin x) + e^x (-\sin x - \cos x)$$

$$= e^x (\cos x - \sin x - \sin x - \cos x)$$

$$= e^x (-2 \sin x)$$

$$f''(x) = -2 e^x \sin x \quad \dots(3)$$

Step 3 : Check whether assumed initial guess is right or wrong

$\therefore$ According to the condition for convergence

$$\frac{f(x_1) \times f''(x_1)}{[f'(x_1)]^2} \leq 1$$

From Equation (1), $f'(x_1) = f(0) = e^0 \cos 0 - 1.2 = -0.20000$

from Equation (2), $f'(x_1) = f'(0) = e^0 (\cos 0) - \sin(0)$

$$f'(x) = 1.0000$$

From Equation (3), $f''(x) = f''(0) = e^0 (-2 \sin 0)$

$$f''(x) = 0$$

$$\therefore \frac{f(x_1) \times f''(x_1)}{[f'(x_1)]^2} = \frac{-0.2 \times 0}{1} = 0$$

$$\therefore 0 < 1$$

Our initial guess $x_1 = 0$ is right.

ITERATION 1

Step 4 : Calculate the new value of x_2 using, Newton Raphson method,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0 - \left(\frac{-0.2}{1} \right) = 0.2000$$

$$\therefore x_2 = 0.20000$$

The remaining iterations are tabulated as below

Iteration no.	x_1	$f(x_1)$	$f'(x_1)$	x_2	Accuracy
1	0	-0.20000	1.00000	0.20000	0.2
2	0.200000	0.002944	0.954401	0.203085	0.0308
3	0.203085	0.000002	0.952690	0.203087	0.00006

The root up to accuracy, 0.001 is 0.203087

Example 2.4.7

Find the positive root of $x^4 - x = 10$ by Newton-Raphson's correct to three decimal places.

Solution :

Given data : $f(x) = x^4 - x - 10 = 0$

Let Initial guess $x_1 = 2$

Accuracy = 0.001

Step 1 : Calculate value of $f'(x)$:

$$\therefore f(x) = x^4 - x - 10 \quad \dots(1)$$

Differentiating above equation w. r. t. x.

$$\therefore f'(x) = 4x^3 - 1 \quad \dots(2)$$

Step 2 : Calculate value of $f''(x)$:

Differentiating equation (2) w.r.to x

$$\therefore f''(x) = 12x^2 \quad \dots(3)$$

Step 3 : Check whether assume initial guess is right or not.

$$\frac{f(x_1) \cdot f''(x_1)}{(f'(x_1))^2} \leq 1$$

from Equation (1) $f(x_1) = 4$

from Equation (2) $f'(x_1) = 4(2)^3 - 1 = 31$

from Equation (3) $f''(x_1) = 12(2)^2 = 48$

$$\therefore \frac{f(x_1) \cdot f''(x_1)}{(f'(x_1))^2} = \frac{4 \times 48}{(31)^2} = 0.199 < 1$$

$$\therefore \frac{f(x_1) \cdot f''(x_1)}{(f'(x_1))^2} < 1$$

$\therefore$ Our assume initial guess x_1 is right.

ITERATION 1

Step 4 : The new value of the root x_2 using Newton - Raphson Method is,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \quad \dots(i)$$

where, $f(x_1) = 2^4 - 2 - 10 = 4$

and $f'(x_1) = 4(2)^3 - 1 = 31$

from Equation (i),

$$\therefore x_2 = 2 - \frac{4}{31} = 1.87$$

Step 5 : Check whether $|x_2 - x_1| < \text{Accuracy}$ or not

$$\therefore |2 - 1.87| = 0.13 > \text{Accuracy}$$

Step 6 : If accuracy is not achieved then replace x_1 by x_2 .

$\therefore$ New initial guess will be $x_1 = x_2$.

or $x_1 = 1.87$

Repeat step 4 to 6 till we get $|x_2 - x_1| < \text{Accuracy}$.

Remaining iterations are tabulated as below.

Iteration number	x_1	$f(x_1)$	$f'(x_1)$	x_2	accuracy
1	2	31	48	1.87	0.13
2	1.87	0.358309	25.156812	1.855781	0.0142
3	1.855781	0.004818	24.564656	1.855585	0.0002

The root correct to 3 decimal places is 1.855585

...Ans.

2.4.4 Stepwise Solution to Solve the Problems on Newton Raphson Method (Iteration Criteria) (Illustrated by Numerical Example)

Example 2.4.8

Determine the root of $f(x) = 0.51x - \sin x$. Using Newton -Raphson Method. Do 3 iterations.

Solution :

Given data : $f(x) = 0.51x - \sin x$

Let Initial guess $x_1 = 2$

Number of Iteration $n = 3$

Step 1 : Calculate value of $f'(x)$:

$$\therefore f(x) = 0.51x - \sin x \quad \dots(1)$$

Differentiating above equation w. r. t. x.

$$\therefore f'(x) = 0.51 - \cos x \quad \dots(2)$$

Step 2 : Calculate value of $f''(x)$:

Differentiating equation (2) w.r.to x

$$\therefore f''(x) = \sin x \quad \dots(3)$$

Step 3 : Check whether assume initial guess is right or not.

$$\frac{f(x_1) \cdot f''(x_1)}{(f'(x_1))^2} \leq 1$$

from Equation (1) $f(x_1) = 0.51(2) - \sin(2) = 0.1107$

from Equation (2) $f'(x_1) = 0.51 - \cos(2) = 0.9261$

from Equation (3) $f''(x_1) = \sin(2) = 0.9092$

$$\therefore \frac{f(x_1) \cdot f''(x_1)}{(f'(x_1))^2} = \frac{0.1107 \times 0.9092}{(0.9261)^2} = 0.1173 < 1$$

$$\therefore \frac{f(x_1) \cdot f''(x_1)}{(f'(x_1))^2} < 1$$

$\therefore$ Our assume initial guess x_1 is right.

ITERATION 1

Step 4 : The new value of the root x_2 using Newton-Raphson Method is,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \quad \dots(i)$$

where, $f(x_1) = 0.51(2) - \sin(2)$
 $= 0.1107$



$$\text{and } f'(x_1) = 0.51 - \cos(2) = 0.9261$$

from Equation (i)

$$\therefore x_2 = 2 - \frac{(0.1107)}{0.9261} = 1.8804$$

Step 5: Replace x_1 by x_2 :

$$\therefore x_1 = x_2 = 1.8804 \quad \text{or } x_1 = 1.8804$$

Repeat the step 4 and 5 for next iteration.

ITERATION 2

Step 4: The new value of the root x_2 using Newton-Raphson Method is,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \quad \dots(ii)$$

where,

$$\begin{aligned} f(x_1) &= f(1.8804) \\ &= 0.51(1.8804) - \sin(1.8804) \\ &= 0.00654 \end{aligned}$$

$$\begin{aligned} \text{and } f'(x_1) &= f'(1.8804) \\ &= 0.51 - \cos(1.8804) = 0.81474 \end{aligned}$$

from Equation (ii)

$$\therefore x_2 = 1.8804 - \frac{0.00660}{0.81474} = 1.87236$$

Step 5: Replace x_1 by x_2

$$\begin{aligned} \therefore x_1 &= x_2 = 1.87236 \\ \text{or } x_1 &= 1.87236 \end{aligned}$$

ITERATION 3

Step 4: The new value of the root x_2 using Newton-Raphson Method is,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \quad \dots(iii)$$

$$\begin{aligned} \text{where, } f(x_1) &= f(1.87236) \\ &= 0.51 \times (1.87236) - \sin(1.87236) \\ &= 0.000031 \end{aligned}$$

$$\begin{aligned} \text{and } f'(x_1) &= f'(1.87236) \\ &= 0.51 - \cos(1.87236) = 0.8070 \end{aligned}$$

$$\begin{aligned} \text{from Equation (iii)} \quad x_2 &= 1.87236 - \left(\frac{0.000031}{0.8070} \right) \\ &= 1.87232 \end{aligned}$$

The result is tabulated as below:

Iteration number	x_1	$f(x_1)$	$f'(x_1)$	x_2
1	2	0.1107	0.9261	1.8804
2	1.8804	0.00660	0.81474	1.87236
3	1.87236	0.000031	0.8070	1.87232

The root of equation after 3 iteration is $x_2 = 1.87232 \dots \text{Ans.}$

Note : Flowchart and Program on Newton Raphson Method (Iteration criteria) are explained in Chapter 9.

Example 2.4.9

Solve the equation $e^x \cdot \cos x - 1.4 \sin x = 0.8$ using Newton-Raphson method taking $x = 1$ and doing 3 iteration

Solution :

$$\text{Given data : } f(x) = e^x \cdot \cos x - 1.4 \sin x - 0.8$$

$$\text{Initial guess } x_1 = 1$$

$$\text{Number of Iteration} = 3$$

Step 1: Calculate value of $f'(x)$:

$$f(x) = e^x \cos x - 1.4 \sin x - 0.8$$

Differentiating above equation w.r. t. x

$$\therefore f'(x) = [e^x \cdot \cos x + e^x (-\sin x)] - 1.4 \cos x - 0$$

$$f'(x) = e^x [\cos x - \sin x] - 1.4 \cos x$$

Skip Step 2 and Step 3 as initial guess is given

$$\text{i.e. } x_1 = 1$$

ITERATION 1

Step 4: The new value of root using Newton-Raphson method:

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\begin{aligned} \text{Now, } f(x_1) &= e^1 \cdot \cos(1) - 1.4 \sin(1) - 0.8 \\ &= -0.5093 \end{aligned}$$

$$\begin{aligned} \text{and } f'(x_1) &= e^1 [\cos(1) - \sin(1)] - 1.4 \cos(1) \\ &= -1.5750 \end{aligned}$$

$$\therefore x_2 = 1 - \left(\frac{-0.5093}{-1.5750} \right)$$

$$\therefore x_2 = 0.6766$$

Step 5: Replace x_1 by x_2 :

$$\therefore \text{New Initial guess } x_1 = x_2 = 0.6766$$

The result is tabulated as below:

Iteration number	x_1	$f(x_1)$	$f'(x_1)$	x_2
2	0.6766	-0.1427	-0.7895	0.4957
3	0.4957	-0.02188	-0.5683	0.45724

The root of equation after 3 iteration is $x_2 = 0.45724 \dots \text{Ans.}$

Example 2.4.10

Using Newton-Raphson Method, and taking initial guess as zero. Find $x^3 - 5x + 3 = 0$. Do 3 iterations only.

Solution :

Given data :

$$f(x) = x^3 - 5x + 3$$

$$\text{Initial guess } x_1 = 0$$

$$\text{Number of Iteration } n = 3$$

Step 1: Find the value of $f'(x)$:

$$\therefore f(x) = x^3 - 5x + 3$$

Differentiating above equation w. r. t. x .

$$\therefore f'(x) = 3x^2 - 5$$

Skip, Step 2 and 3 as the Initial guess $x_1 = 0$ is given.

ITERATION 1

Step 4: The new value of the root x_2 , using Newton-Raphson Method is:

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\text{where, } f(x_1) = (0)^3 - 5(0) + 3 = 3$$

$$\text{and } f'(x_1) = 3(0)^2 - 5 = -5$$

$$\therefore x_2 = 0 - \left(\frac{3}{-5}\right)$$

$$x_2 = 0.6$$

Step 5: Replace value of x_1 by x_2

$$\therefore x_1 = x_2 = 0.6$$

New guess $x_1 = 0.6$

The remaining iterations are tabulated as below:

Iteration number	x_1	$f(x_1)$	$f'(x_1)$	x_2
2	0.6	0.216	-3.92	0.6551
3	0.6551	0.0056	-3.7125	0.6566

The root of the equation after 3 iteration is

$$x_2 = 0.6566 \quad \dots \text{Ans.}$$

Example 2.4.11

Solve $e^x \cdot \cos x - 1.2 = 0$ by Newton-Raphson Method. Take initial guess $x = 1.2$. Make at least four iteration.

Solution:

$$\text{Given data: } y = f(x) = e^x \cdot \cos x - 1.2$$

$$\text{Initial guess } x_1 = 1.2$$

$$\text{Number of Iteration } n = 4$$

Step 1: Calculate value of $f'(x)$:

$$\therefore f(x) = e^x \cdot \cos x - 1.2$$

Differentiating above equation w. r. t. x .

$$\begin{aligned} \therefore f'(x_1) &= e^x \cdot \cos x + e^x \cdot (-\sin x) - 0 \\ &= e^x [\cos x - \sin x] \end{aligned}$$

Skip, Step 2 and 3 as the initial guess $x_1 = 1.2$ is given

ITERATION 1

Step 4: The new value of the root x_2 using Newton-Raphson Method is

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\begin{aligned} \text{where, } f(x_1) &= e^{1.2} \cdot \cos(1.2) - 1.2 \\ &= 0.00301 \end{aligned}$$

$$\text{and } f'(x_1) = e^{1.2} [\cos(1.2) - \sin(1.2)]$$

$$= -1.8914$$

$$\therefore x_2 = 1.2 - \left(\frac{0.00301}{-1.8914}\right)$$

$$\therefore x_2 = 1.20162$$

Step 5: Replace x_1 by x_2 :

$$\therefore x_1 = x_2 = 1.20162$$

$\therefore$ The new guess $x_1 = 1.20162$.

The remaining iterations are tabulated as below.

Iteration number	x_1	$f(x_1)$	$f'(x_1)$	x_2
2	1.20162	0.00359	-1.911	1.2034
3	1.2034	0.00356	-1.9125	1.20526
4	1.20526	0.00697	-1.924	1.20163

The root of the equation after 4 iteration is,

$$x_2 = 1.20163 \quad \dots \text{Ans.}$$

Example 2.4.12

Find a positive real root of $3x = \cos x + 1$ by Newton's method.

Solution:

$$\text{Given data: } f(x) = \cos x - 3x + 1 \quad \dots (1)$$

Let initial guesses $x_1 = 0$

Step 1: Calculate value of $f'(x)$

Differentiate Equation (1) w.r. to x

$$\therefore f'(x) = -\sin x - 3 \quad \dots (2)$$

Step 2: Calculate value of $f''(x)$

Differentiate Equation (2) w.r. to x

$$\therefore f''(x) = -\cos x \quad \dots (3)$$

Step 3: Check whether assume initial guess is right or wrong.

$$\frac{f(x_1) \cdot f''(x_1)}{(f'(x_1))^2} \leq 1$$

$$\text{From Equation (1) } f(x_1) = \cos(0) - (3 \times 0) + 1 = 2$$

$$\text{From Equation (2) } f'(x_1) = -\sin(0) - 3 = -3$$

$$\text{From Equation (3) } f''(x_1) = -\cos(0) = -1$$

$$\therefore \frac{f(x_1) \cdot f''(x_1)}{(f'(x_1))^2} = \frac{2 \times (-1)}{(-3)^2} = -0.222 \leq 1$$

$\therefore$ Our assume initial guess x_1 is right

ITERATION 1

Step 4: The new value of root x_2 using Newton's Raphson method is,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0 - \frac{2}{-3} = 0.6666$$

Step 5: Replace x_1 by x_2

$$\therefore x_1 = x_2 = 0.6666$$

$$\text{or } x_1 = 0.6666$$

Repeat step 4 and 5 till number of iteration

**ITERATION 2**

Step 4: The new value of root x_2 using Newton's Raphson method

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \quad \dots(i)$$

$$\text{Where } f(x_1) = \cos(0.6666) - (3 \times 0.6666) + 1 \\ = -0.2138$$

$$f'(x_1) = -\sin(0.6666) - 3 = -3.6183$$

From Equation (i)

$$x_2 = 0.6666 - \frac{1.3232}{-3.0116} = 0.6075$$

Step 5: Replace x_1 by x_2

$$\therefore x_1 = x_2 = 0.6075 \text{ or } x_1 = 0.6075$$

Remaining iterations are tabulated as below

Iteration No.	x_1	$f(x_1)$	$f'(x_1)$	x_2
1	0	2	-3	0.6666
2	0.6666	-0.2138	-3.6183	0.6075
3	0.6075	-0.001422	-3.5708	0.6071
4	0.6071	0.005884	-3.5704	0.6087
5	0.6087	-0.005707	-3.5718	0.6071
6	0.6071	0.005884	-3.5704	0.6087

The root of equation upto 6 iteration is 0.6087

Syllabus Topic : Engineering Applications**2.5 Engineering Applications**

1. An engineers, use the following formula to calculate the upward velocity of the rocket,

$$v = u \log \left(\frac{m_0}{m_0 - qt} \right) - gt$$

Where, v = upward velocity,

u = velocity at which fuel is expelled relative to rocket,

m_0 = Initial mass of the rocket at time $t = 0$

q = fuel consumption rate and

g = acceleration due to gravity,

Now if we want to calculate time, if the values of u , m_0 , q and v are known then equation in terms of " t " can be written as,

$$f(t) = u \log \left(\frac{m_0}{m_0 - qt} \right) - (gt + v).$$

Therefore the value of " t " that makes the function $f(t) = 0$ is called the root of the function. Therefore, we can calculate value of " t " at a particular given data using iterative method.

2. A car covers certain distance with some speed. If its speed is increased by a certain km/hr, it will cover the same distance in particular hour less. Then we can find the algebraic equation governing the speed of the car as

$$\text{Speed of car} = \frac{\text{Distance travelled}}{\text{time}}$$

$$\therefore v = \frac{s}{t}$$

$$\therefore s = v \times t$$

where (s in km, v in km/hr and t in hr)

Then the equation for speed of car.

$$f(v) = \text{Equation in terms of } v$$

and the speed of the car can be calculated using iterative method.

2.5.1 Engineering Applications Problems**Example 2.5.1**

A car covers 310 km with some speed. If its speed is increased by 10 km/hr, it will cover the same distance in one hour less. Find out the algebraic equation governing the speed of the car and find the speed of the car by bi-section method.

Solution :

Given data : Distance Travelled = 310 km.

$$\text{Speed of car } v = \frac{\text{Distance travelled}}{\text{time}}$$

$$\therefore v = \frac{s}{t}$$

$\therefore s = v \times t$ where (s in km, v in km/hr and t in hr)

For 1st condition, when $s_1 = 310$ km for speed of v_1 and for time t_1

$$\therefore s_1 = v_1 \times t_1$$

$$\therefore 310 = v_1 \times t_1 \quad \dots(1)$$

$$\therefore t_1 = 310/v_1$$

for 2nd condition, $s_1 = 310$ km for speed of $(v_1 + 10)$ and for time $(t_1 - 1)$

$$\therefore s_1 = (v_1 + 10) \times (t_1 - 1)$$

$$\therefore s_1 = v_1 t_1 - v_1 + 10 t_1 - 10$$

$$\text{But } s_1 = s_1 - v_1 + 10 t_1 - 10$$

$$\therefore v_1 = 10 t_1 - 10$$

$$\therefore t_1 = \left(\frac{v_1 + 10}{10} \right) \quad \dots(2)$$

compare Equation (1) and (2)

$$\therefore \frac{310}{v_1} = \frac{v_1 + 10}{10}$$

$$\therefore v_1^2 + 10 v_1 - 3100 = 0$$

This is the required governing equation for speed of car.

$$\text{Let } f(v) = v^2 + 10 v - 3100$$

Assume initial guesses as $v_1 = 50$

$$\text{and } v_2 = 55$$

Step 1 : Check whether initial guesses are right or wrong.

$$\text{For } v_1 = 50$$

$$f(v_1) = 50^2 + 10 \times 50 - 3100 \\ = -100 = \text{negative}$$

$$\text{For } v_2 = 55$$

$$\therefore f(v_2) = 55^2 + 10 \times 55 - 3100 \\ = 475 = \text{positive}$$

$\therefore$ Now $f(v_1) \times f(v_2) < 0$ i.e. (both have opposite signs)

Hence the initial guesses v_1 and v_2 are right.

$$\therefore v_1 = 50, v_2 = 55 \text{ and}$$

$$f(v_1) = -100, f(v_2) = 475$$



ITERATION 1

Step 2: Using bi-section method.

$$v_3 = \frac{v_1 + v_2}{2} = \frac{50 + 55}{2} = 52.5$$

Step 3: Check whether $f(v_3)$ is either positive or negative.

$$\begin{aligned} \therefore f(v_3) &= (52.5)^2 + 10(52.5) - 3100 \\ &= 181.25 = \text{positive} \end{aligned}$$

Step 4: Replace v_1 or v_2 by v_3

$$\begin{aligned} \therefore f(v_1) &= \text{positive}, f(v_2) = \text{positive}, \\ f(v_3) &= \text{positive} \end{aligned}$$

$$\therefore v_2 = v_3 = 52.5 \text{ and } f(v_2) = f(v_3) = 181.25$$

Repeat step 2 to 4 for next iterations

ITERATION 2

Step 2: Using bi-section method.

$$\therefore v_3 = \frac{v_1 + v_2}{2} = \frac{50 + 52.5}{2} = 57.25$$

Step 3: Check whether $F(V_3)$ is positive or negative.

$$\begin{aligned} \therefore f(v_3) &= (57.25)^2 + 10(57.25) - 3100 \\ &= 39.06 = \text{positive} \end{aligned}$$

Step 4: Replace v_1 or v_2 by v_3

$$\therefore f(v_1) = \text{negative}, f(v_2) = \text{positive}, f(v_3) = \text{positive}$$

$$\therefore v_1 = v_3 = 57.25$$

$$\text{and } f(v_1) = f(v_3) = 39.06$$

The results are tabulated as below:

Iteration number	v_1	$f(v_1)$	v_2	$f(v_2)$	v_3	$f(v_3)$
1	50	-100	55	475	52.5	181.25
2	50	-100	52.5	181.25	51.25	39.08

After two iterations speed of the car

$$v_3 = 51.25 \text{ km/hr.}$$

...Ans.

Example 2.5.2

The vander waal's equation for real gases is given by,

$$\left(p + \frac{a}{v^2}\right)(v - b) = RT$$

Where

p = pressure = 1 kN/m², R = Gas constant = 0.082 kJ/kg K, a = constant = 3.82 and b = constant = 0.06 and v = volume at pressure p and T = Temperature in K. Find the volume at 300 K. Do two iterations. Assume Initial volume = 20 m³/kg.

Solution:

$$\text{Given data: } f(v) = \left(p + \frac{a}{v^2}\right)(v - b) - RT = 0$$

Substituting $p = 1$, $R = 0.082$, $a = 3.82$, $b = 0.06$ and $T = 300$ in the above equation $f(v)$ becomes.

$$\therefore f(v) = \left(1 + \frac{3.82}{v^2}\right)(v - 0.06) - 24.6 \quad \dots(i)$$

Number of Iteration = 2.

Step 1: Find the value of $f'(v)$:

Differentiating Equation (i) w. r. t. x .

$$\therefore f'(v) = \left(1 + \frac{3.82}{v^2}\right)(1) + (v - 0.06) \times \left(\frac{-2 \times 3.82}{v^3}\right)$$

$$= 1 + \frac{3.82}{v^2} - \frac{7}{v^2} + \frac{0.06 \times 7}{v^3}$$

$$f'(v) = 1 - \frac{3.82}{v^2} + \frac{0.4584}{v^3}$$

Skip step 2 and 3 as the Initial guess $v_1 = 20$ is given.

ITERATION 1

Step 4: The new value of v_2 using Newton - Raphson Method is,

$$v_2 = v_1 - \frac{f(v_1)}{f'(v_1)} \quad \dots(ii)$$

where, $f(v_1) = f(20)$

$$= \left[1 + \frac{3.82}{20^2}\right](20 - 0.06) - 24.6$$

$$f(v) = -4.469$$

$$\text{and } f'(v_1) = f'(20) = \left[1 - \frac{3.82}{20^2}\right] + \frac{-0.4584}{20^3}$$

$$f'(v) = 0.990$$

Substituting in the Equation (ii),

$$v_2 = 20 - \left(\frac{-4.469}{0.990}\right) = 24.512$$

Step 5: Replace v_1 by v_2

$$\therefore v_1 = v_2 = 24.512 \text{ m}^3/\text{kg}$$

Repeat step 4 and 5 for next iteration

ITERATION 2

Step 4: The new value of v_2 using Newton - Raphson Method is,

$$v_2 = v_1 - \frac{f(v_1)}{f'(v_1)} \quad \dots(iii)$$

where, $f(v_1) = f(24.512)$

$$= \left[1 + \frac{3.82}{24.512^2}\right](24.512 - 0.06) - 24.6$$

$$f(v) = 0.007865$$

$$\text{and } f'(v_1) = f'(24.512)$$

$$= \left[1 - \frac{3.82}{24.512^2}\right] + \frac{-0.4584}{24.512^3}$$

$$f'(v) = 0.9936$$

Substituting in the Equation (ii),

$$v_2 = 24.512 - \left(\frac{0.007865}{0.9936}\right) = 24.50$$

The other iteration result is tabulated as below:

Iteration number	v_1	$f(v_1)$	$f'(v_1)$	v_2
1	20	-4.46	0.990	24.512
2	24.512	0.00786	0.9936	24.50

$\therefore$ After 2 iterations

The volume at 300° K is $v_2 = 24.50 \text{ m}^3/\text{kg}$

...Ans.

Example 2.5.3

The bacteria concentration (C), in a reservoir varies as $C = 4e^{-2t} + e^{-0.1t}$. Using Newton-Raphson method, calculate the time required for the bacteria concentration to be 0.5 accurate upto 4 decimal places.

Solution :

Given data : Function $c = f(t)$

$$c = 4e^{-2t} + e^{-0.1t}$$

$$c = 0.5$$

$$\therefore 4e^{-2t} + e^{-0.1t} = 0.5 = 0$$

Let initial guess $t_1 = 10$

Step 1 : Calculate the value of $f'(t)$

$$\text{As } f(t) = 4e^{-2t} + e^{-0.1t}$$

$$f(t) = -0.13212$$

$$\therefore f'(t) = -0.03679$$

Step 2 : Calculate the value of $f''(t)$

$$\therefore f''(t) = 3.67883 \times 10^{-3}$$

Step 3 : Check whether assume guess is right or wrong.

$$\text{As } \frac{f(t)f''(t)}{(f'(t))^2} = -0.3591 < 1$$

$$\therefore t_1 = 10 \text{ is right}$$

ITERATION 1

Step 4 : The new value of root t_2 by Newton Raphson method is

$$\begin{aligned} t_2 &= t_1 - \frac{f(t_1)}{f'(t_1)} \\ &= 10 - \left(\frac{-0.13212}{-0.03679} \right) \\ t_2 &= 6.408806 \end{aligned}$$

Step 5 : Check whether $|t_2 - t_1| < \text{accuracy}$ or not.

$$|t_2 - t_1| = |6.408806 - 10| = 3.59119 < 0.0001$$

Step 6 : If accuracy is not achieved then, replace t_1 by t_2 .

Replace t_1 by t_2

$$\therefore t_1 = t_2 = 6.408806$$

The other iterations result are tabulated below :

Iteration number	t_1	$f(t_1)$	$f'(t_1)$	t_2	Accuracy
1	10	-0.13212	-0.03679	6.408806	3.59119
2	6.408806	0.02684	-0.0527	6.91806	0.50925
3	6.91806	6.74959×10^{-4}	-0.5007	6.93154	0.01348
4	6.93154	4.04469×10^{-7}	-0.0500072	6.931548	8.0882×10^{-5}

$\therefore$ The root upto acceleration of 0.0001 is

$$t_2 = 6.931548 \text{ sec}$$

Example 2.5.4

Find the drag coefficient C for a mass $m = 73.5 \text{ kg}$ to have a velocity of 32.8 m/s after falling for time $t = 10 \text{ sec}$.

The acceleration due to gravity is 9.8 m/s^2 .

Newton's second law equation for velocity is given by,

$$V = \frac{gm}{c} [1 - e^{-(c/m)t}]$$

Solve the problem by numerical method.

Solution :

Given data : $V = \frac{gm}{c} [1 - e^{-(c/m)t}]$

$\therefore$ To find drag coefficient C, The function in terms of $f(c)$ is given by,

$$f(c) = \frac{gm}{c} [1 - e^{-(c/m)t}] - v$$

$$\therefore f(c) = \frac{9.8 \times 73.5}{c} [1 - e^{-(c/73.5) \cdot 10}] - 32.8$$

$$\text{Let } y = f(c) = \frac{720.3}{c} [1 - e^{-0.136c}] - 32.8$$

Now the problem can be solved by using any numerical method.

Example 2.5.5

The charge dissipated of a electrical circuit is given by,

$$\frac{q}{q_0} = e^{-Rt/2L} \cos \left[\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2} t \right]$$

Where, $\frac{q}{q_0} = 0.04$, R = Resistance in Ω ,

L = Inductance = 6 H , t = time = 0.08 sec ,

C = capacitance = 10^{-4} F . Calculate the resistance R using numerical method.

Solution :

Given data : $\frac{q}{q_0} = e^{-Rt/2L} \cos \left[\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2} t \right]$

$\therefore$ To find the resistance R , the function in terms of $f(R)$ is given by.

$$f(R) = e^{-R/2L} \cdot \cos \left[\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2} t \right] - \frac{q}{q_0}$$

Substituting the value of $\frac{q}{q_0}$, L , t and C we get,

$$f(R) = e^{-0.006 R} \cdot \cos (\sqrt{1666.66 - 0.00694 R^2} \times (0.08)) - 0.04$$

Now the problem can be solved using any numerical method.

Example 2.5.6

The upward velocity of a rocket can be computed by the following data $v = u \log \left(\frac{m_0}{m_0 - qt} \right) - gt$

Where v is the upward velocity, u is the velocity at which fuel is expelled, m_0 is the initial mass of the rocket at time t , q is the fuel consumption rats, and $g = 9.8 \text{ m/s}^2$

If $u = 1800 \text{ m/s}$, $m_0 = 150,000 \text{ kg}$, $q = 2400 \text{ kg/sec}$. Compute the time at which $v = 1000 \text{ m/s}$ by numerical method. Take Initial guesses between $t = 10 \text{ sec}$ and $t = 50 \text{ sec}$.

Solution :

Given data :

$$\text{Let } y = f(t) = u \log \left(\frac{m_0}{m_0 - qt} \right) - gt - v$$

Where $v = 1000 \text{ m/s}$, $u = 1800 \text{ m/s}$, $m_0 = 150,000 \text{ kg}$,

$$q = 2400 \text{ kg/sec}, g = 9.8 \text{ m/s}^2$$

Substituting all the values in above equation,

$$y = f(t) = 1800 \log \left(\frac{150000}{150000 - 2400 \times t} \right) - 9.8 t - 1000$$

Now the problem can be solved using any numerical method.

...Chapter Ends

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CHAPTER 3 UNIT III

Numerical Solutions of Algebraic Equations

Syllabus :

Motivation, Cramer's rule, Gauss- Elimination Method, pivoting, scaling, engineering applications.

Syllabus Topic : Motivation

3.1 Introduction

One of the most fundamental problems in Engineering and Science is that of solving a set of simultaneous linear algebraic equations. These linear equations are used in variety of applications in the field of Science and Engineering.

A system of linear equations with 'n' unknowns is represented as,

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$a_{31}x_1 + a_{32}x_2 + \dots + a_{3n}x_n = b_3$$

$$\dots\dots\dots$$

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n$$

$x_1, x_2, x_3, \dots, x_n$ are the variables whose values are to be found.

A set of equations is known as system of simultaneous equations.

In matrix format, the above equation can be expressed as

$$AX = B$$

Where $A = n \times n$ coefficient matrix ; b is an $n \times 1$ right hand side vector, and x is an $n \times 1$ vector of unknowns.

The objective is to find a value of x which satisfies all the equations.

3.1.1 Various Available Solutions

While solving system of equations, we come across the four types of solutions. They are,

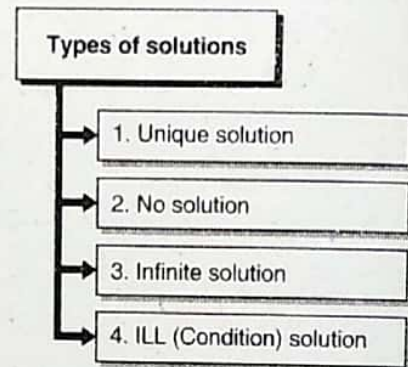


Fig. C3.1 : Types of solutions

→ 1. Unique Solution

When a system has only one solution, then the system is referred as a system with Unique Solution.

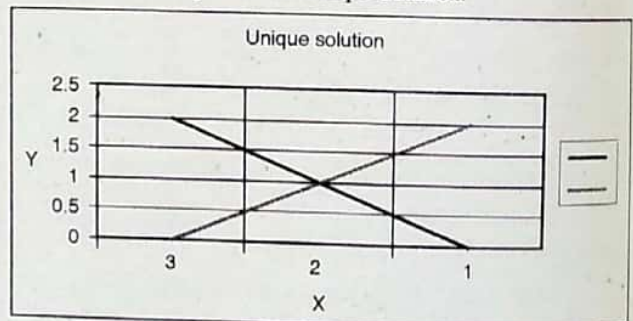


Fig. 3.1.1

Consider equation,

$$x + y = 3 \quad \dots(1)$$

$$x - y = 1 \quad \dots(2)$$

∴ Adding Equation (1) and (2), we get,

$$x = 2 \text{ and } y = 1.$$

The above system has a solution as $x = 2$ and $y = 1$ which satisfy the above two equations. No other values of x and y can

satisfy the above two equation. Hence, the system have unique solution as $x = 2$ and $y = 1$.

From the Fig. 3.1.1 as two line intersects at only one point hence the system has unique solution.

→ 2. No Solution

If the system doesn't have any solution, then the system is referred as a system with No Solution.

Consider equation,

$$x + y = 3 \quad \dots(1)$$

$$x + y = 1 \quad \dots(2)$$

Subtracting equation 1 and 2 we can't find value of x and y . Hence no value x and y can satisfy the above two equation. Hence, the system has no solution.

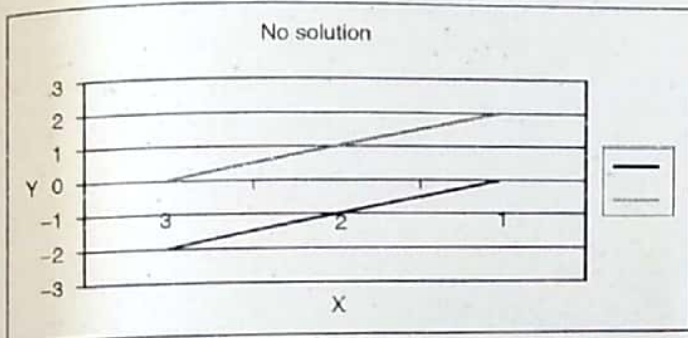


Fig. 3.1.2

As the two lines are parallel and do not intersect each other at any point, hence the system has no solution.

→ 3. Infinite Solution

If the system has more than one solution, then the system is referred as a system with infinite solution.

Consider equation,

$$-x + y = 2 \quad \dots(1)$$

$$2x - 2y = -4 \quad \dots(2)$$

On solving equation 1 and 2, we can say that, more than one value of x and y can satisfy the above two equations.

i.e. At $x = 3$ and $y = 5$, the values of x and y satisfy both equation and also At $x = 1$ and $y = 3$, the values of x and y satisfy both equation

i.e. At $x = 4$ and $y = 6$, the values of x and y satisfy both equation

As the two lines coincides with one another, hence they satisfy various points of x and y . Hence, the system has many solution.

→ 4. Ill (condition) Solution

If the system has a solution which is very very close to being unique, then the system is referred as a system with ill (condition) solution.

Consider the equation

$$x + y = 2$$

$$0.482x - 0.92y = 1.6$$

Ill conditioned systems are those where small changes in the coefficients results in large changes in the solution.

An alternate interpretation of ill conditioning is that a wide range of answer can approximately satisfy the equation.

In order to have to unique solution, the number of equation should be equal to number of unknown.

To have the solution, we have to convert the linear equation into matrix form and then perform various operations on the matrix, so we have to study and revise few topics of the matrix first.

3.2 Introduction to Matrix

An arrangement of numbers in rows and columns is called matrix.

$$\text{e.g. } A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 5 & 9 \end{bmatrix}_{2 \times 3}$$

There are 2 rows and 3 columns. This is a rectangular matrix.

Types of Matrices

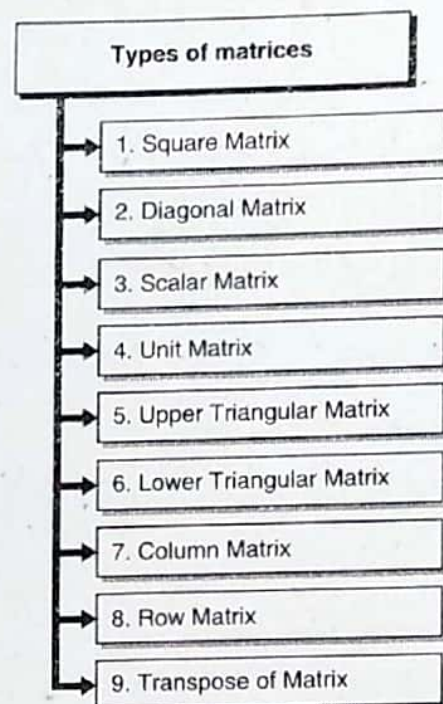


Fig. C3.2 : Types of matrices

→ 1. Square Matrix

It is a matrix in which Number of rows = Number of columns. e.g. 2×2 , 3×3 etc.

→ 2. Diagonal Matrix

It is a square matrix in which all non-diagonal elements are zero.

$$\text{e.g. } A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}_{4 \times 4} \quad \text{or } A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

→ 3. Scalar Matrix

It is a diagonal matrix in which all diagonal elements are equal and all non-diagonal elements are zero.

$$\text{e.g. } A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}_{3 \times 3}$$

**→ 4. Unit Matrix**

It is a scalar matrix with diagonal elements as unity.

$$\text{e.g. } I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

Note : For any matrix $A = AI = IA = A$

→ 5. Upper Triangular Matrix

It is a square matrix in which all the elements below the principal diagonal are zero.

$$\text{e.g. } A = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 2 & 1 \\ 0 & 0 & 5 \end{bmatrix}_{3 \times 3}$$

→ 6. Lower Triangular Matrix

It is a square matrix in which all elements above principle

$$\text{diagonal are zero. } \begin{bmatrix} 1 & 0 & 0 \\ 3 & 4 & 0 \\ -1 & 3 & 2 \end{bmatrix}_{3 \times 3}$$

→ 7. Column Matrix

It is a matrix in which there is only one column.

$$\text{e.g. } X = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}_{3 \times 1}$$

→ 8. Row Matrix

It is a matrix in which there is only one row.

$$\text{e.g. } X = [5 \ 7 \ 9]_{1 \times 3}$$

Note : 1. Row and column matrices are called as vectors.

2. The entries in a matrix are called as scalars.

→ 9. Transpose of Matrix

It is a matrix obtained by interchanging rows into columns (or columns into rows).

$$\text{e.g. } A = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 7 & 9 \end{bmatrix}_{2 \times 3}$$

$$A' = \text{Transpose of } A = \begin{bmatrix} 1 & 3 \\ 3 & 7 \\ 5 & 9 \end{bmatrix}_{3 \times 2}$$

3.3 Homogeneous and Non-Homogeneous Equation

Consider a set of equations :

$$a_1 x + b_1 y + c_1 z = d_1$$

$$a_2 x + b_2 y + c_2 z = d_2$$

$$a_3 x + b_3 y + c_3 z = d_3$$

The equations can be written in the matrix form as :

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$A \quad X \quad D$$

$$\text{i.e. } A \cdot X = D$$

If the column matrix, D is non-zero matrix, the system is said to be Non-Homogeneous. If it is zero matrix, then the system is Homogeneous.

Now we join matrices A and D as :

$$[A : D] = \begin{bmatrix} a_1 & b_1 & c_1 & : & d_1 \\ a_2 & b_2 & c_2 & : & d_2 \\ a_3 & b_3 & c_3 & : & d_3 \end{bmatrix}$$

It is called augmented matrix.

Consider a set of homogeneous equations, (In a homogeneous equation, constant term is zero or matrix D is a zero matrix and denoted by O.)

$$a_1 x + b_1 y + c_1 z = 0$$

$$a_2 x + b_2 y + c_2 z = 0$$

$$a_3 x + b_3 y + c_3 z = 0$$

Note :

(1) Homogeneous set of equations is always consistent, since it has always atleast one solution viz. $x_1 = x_2 = x_3 = 0$. This solution is called as Trivial solution or Zero solution or Unique solution.

(2) The system may possess infinitely many solutions. These solutions are called as Non-trivial or Non-zero or infinitely many solutions.

(3) There is a method of finding the nature of equations and it is as follows :

(i) If $|A| \neq 0$

(ii) and if $|A| = 0$ then the system has non-zero solutions.

3.4 Various Methods to Solve Linear Algebraic Equations

The solution of system of Linear algebraic equation could be classified into three categories.

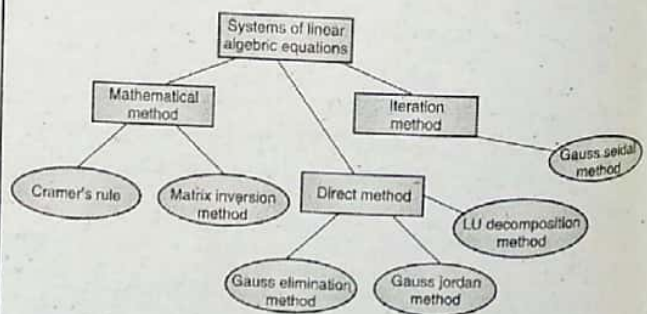


Fig. 3.4.1

**1. Mathematical Method**

- Gamer's rule
- Matrix inversion method

2. Direct Method

In which the elimination approach is taken to achieve the solution. The solution obtained is exact solution.

The various direct methods are :

- Gauss Elimination method
- Gauss Jordan Method
- LU Decomposition Method.
- Cholesky Method.

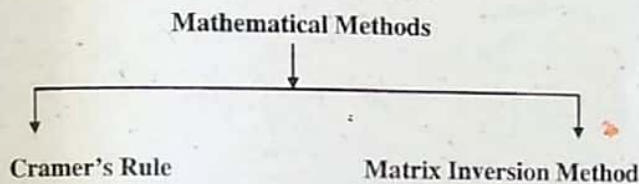
3. Iterative Method

In which the iterative approach is taken to achieve the solution. The solution obtained is approximate solution. The various indirect methods are

- Gauss Siedal Method
- Jacobi's Method

3.5 Mathematical Method

Mathematical Methods are classified as :

**Syllabus Topic : Cramer's Rule****3.5.1 Cramer's rule**

The system of equations for three unknown variables is :

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Then in the above equations x_1 , x_2 and x_3 are the unknown variables which have to be found.

$$x_1 = \frac{\Delta x_1}{\Delta}, \quad x_2 = \frac{\Delta x_2}{\Delta}, \quad x_3 = \frac{\Delta x_3}{\Delta}$$

where Δ is the determinant of coefficients,

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\therefore \Delta = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$= a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22})$$

The value of Δx_1 is obtained by replacing first column in Δ by b_1 , b_2 and b_3 , i.e.

$$\Delta x_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}$$

$$\therefore \Delta x_1 = b_1 \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} b_2 & a_{23} \\ b_3 & a_{23} \end{vmatrix} + a_{13} \begin{vmatrix} b_2 & a_{23} \\ a_3 & a_{32} \end{vmatrix}$$

$$= b_1(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(b_2a_{33} - b_3a_{23}) + a_{13}(b_2a_{32} - b_3a_{22})$$

The value of Δx_2 is obtained by replacing second column in Δ by b_1 , b_2 and b_3 i.e.

$$\Delta x_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}$$

$$\therefore \Delta x_2 = a_{11} \begin{vmatrix} b_2 & b_{23} \\ b_3 & a_{33} \end{vmatrix} - b_1 \begin{vmatrix} a_{21} & b_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & b_2 \\ a_{31} & b_3 \end{vmatrix}$$

$$\Delta x_2 = a_{11}(b_2a_{33} - b_3a_{23}) - b_1(a_{21}a_{33} - a_{31}b_{23}) + a_{13}(a_{21}b_2 - a_{31}b_2)$$

Similarly Δx_3 can be obtained by replacing third column in Δ by b_1 , b_2 and b_3 .

$$\text{i.e. } \Delta x_3 = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$$

$$\text{and similarly, } \Delta x_3 = a_{11}(a_{22}b_3 - a_{32}b_2) - a_{12}(a_{21}b_3 - a_{31}b_2) + b_1(a_{21}a_{32} - a_{31}a_{22})$$

then we can find the value of x_1 , x_2 , x_3 by the following equations :

$$x_1 = \frac{\Delta x_1}{\Delta}, \quad x_2 = \frac{\Delta x_2}{\Delta}, \quad \text{and } x_3 = \frac{\Delta x_3}{\Delta}$$

STEPWISE SOLUTION TO SOLVE THE PROBLEMS**ON CRAMER'S RULE**

(Illustrated by Numerical Example)

Example 3.5.1

Solve the following linear simultaneous equation by any suitable method.

$$3.0x - 0.1y - 0.2z = 7.85$$

$$0.1x + 7.0y - 0.3z = -19.30$$

$$0.3x - 0.2y + 10.0z = 71.40$$



$$\begin{aligned}
 &= 0.3 [(1 \times -0.44) - (0.3 \times 0.67)] - 0.52 [(0.5 \times -0.44) \\
 &\quad - (0.1 \times 0.67)] - 0.01 [(0.5 \times 0.3) - (0.1 \times 1)] \\
 &= 0.3 (-0.641) - 0.52 (-0.287) - 0.01 (0.005) \\
 &= -0.004356
 \end{aligned}$$

Step 5: Find the value of x_1, x_2, x_3 .

$$\therefore x_1 = \frac{\Delta x_1}{\Delta} = \frac{0.03278}{0.0022} = -14.9$$

$$x_2 = \frac{\Delta x_2}{\Delta} = \frac{0.0639}{0.0022} = -29.5$$

$$x_3 = \frac{\Delta x_3}{\Delta} = \frac{-0.04356}{-0.0022} = 19.8$$

$$\therefore x_1 = -14.9, \quad x_2 = -29.5, \quad x_3 = 19.8 \quad \dots \text{Ans.}$$

Example 3.5.3

Solve by Cramer's rule :

$$x - 3y + z = 2$$

$$3x + y + z = 6$$

$$5x + y + 3z = 3$$

Solution :

Step 1 : Find the value of Δ .

$$\text{Let } \Delta = \begin{vmatrix} 1 & -3 & 1 \\ 3 & 1 & 1 \\ 5 & 1 & 3 \end{vmatrix} = 12.$$

Step 2 : Find the value of Δx .

$$\Delta x = \begin{vmatrix} 2 & -3 & 1 \\ 6 & 1 & 1 \\ 3 & 1 & 3 \end{vmatrix} = 52$$

Step 3 : Find the value of Δy .

$$\Delta y = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 6 & 1 \\ 5 & 3 & 3 \end{vmatrix} = -14$$

Step 4 : Find the value of Δz .

$$\Delta z = \begin{vmatrix} 1 & -3 & 2 \\ 3 & 1 & 6 \\ 5 & 1 & 3 \end{vmatrix} = 70$$

Step 5 : Find the value of x, y, z .

$$\text{Hence, } x = \frac{\Delta x}{\Delta} = \frac{52}{12} = \frac{13}{3}$$

$$y = \frac{\Delta y}{\Delta} = \frac{-14}{12} = -\frac{7}{6}$$

$$z = \frac{\Delta z}{\Delta} = \frac{70}{12} = \frac{35}{6}$$

$$\therefore x = \frac{13}{3}, y = -\frac{7}{6}, z = \frac{35}{6} \quad \dots \text{Ans.}$$

Syllabus Topic : Gauss Elimination Methods

3.6 Substitution Methods for Gauss Elimination

When the number of equations increases, the direct methods like Cramer's rule and Matrix Inversion method become very complex. Then Substitution method is used for large number of equations.

3.6.1 Backward Substitution Method

Consider the following equations involving four unknowns.

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 = b_1 \quad \dots(1)$$

$$0 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 = b_2 \quad \dots(2)$$

$$0 + 0 + a_{33}x_3 + a_{34}x_4 = b_3 \quad \dots(3)$$

$$0 + 0 + 0 + a_{44}x_4 = b_4 \quad \dots(4)$$

In the matrix form $AX = B$, equation can be written as,

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

We can see that from Equation (4) we can find x_4

$$x_4 = \frac{b_4}{a_{44}}$$

Substitute for x_4 in Equation (3) we get,

$$a_{33}x_3 + a_{34}\frac{b_4}{a_{44}} = b_3$$

$$\therefore x_3 = \frac{1}{a_{33}} \left(b_3 - a_{34}\frac{b_4}{a_{44}} \right)$$

Now we can substitute x_4 and x_3 in Equation (2) to get x_2

Finally we can substitute x_4, x_3 and x_2 in Equation (1) to get x_1 .

3.6.2 Forward Substitution Form

Consider a set of equations in lower triangular form :

$$a_{11}x_1 + 0 + 0 + 0 = b_1 \quad \dots(1)$$

$$a_{21}x_1 + a_{22}x_2 + 0 + 0 = b_2 \quad \dots(2)$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + 0 = b_3 \quad \dots(3)$$

$$a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + a_{44}x_4 = b_4 \quad \dots(4)$$

In the matrix form $AX = B$, the equation can be written as,

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

We start from Equation (1) i.e.

$$a_{11}x_1 = b_1$$

$$\therefore x_1 = \frac{b_1}{a_{11}}$$

Equation (2) is, $a_{21}x_1 + a_{22}x_2 = b_2$



Substitute the value of x_1 in this equation,

$$a_{21} \cdot \frac{b_1}{a_{11}} + a_{22} x_2 = b_2$$

$$\therefore x_2 = \frac{1}{a_{22}} \left(b_2 - a_{21} \cdot \frac{b_1}{a_{11}} \right) \quad \dots(5)$$

Similarly we can obtain the value of x_3 from Equation (3) by putting values of x_1 and x_2 and finally we can substitute x_1 , x_2 and x_3 in Equation (4) to get value of x_4 .

3.7 Gauss Elimination Method

Review Questions

- Q. Write step by step procedure for Gauss elimination method.
- Q. When does the Gauss elimination method fail? Explain.

This method is employed in two steps :

- Forward Elimination and
- Backward Substitution.

The method is described as below, where three equations are considered for simplicity.

The equations in general form are,

$$a_{11} x_1 + a_{12} x_2 + a_{13} x_3 = c_1$$

$$a_{21} x_1 + a_{22} x_2 + a_{23} x_3 = c_2$$

$$a_{31} x_1 + a_{32} x_2 + a_{33} x_3 = c_3$$

In the matrix form $AX = B$ the equations can be written as,

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

The matrix of coefficients is reduced to upper triangular form by using **row transformations**. Thus the matrix changes to,

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22}' & a_{23}' \\ 0 & 0 & a_{33}'' \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} c_1' \\ c_2' \\ c_3'' \end{bmatrix}$$

This procedure is called as Forward Elimination. The values of ' x_1 ' are now obtained by Backward Substitution as,

$$x_3 = \frac{c_3''}{a_{33}''}; x_2 = \frac{(c_2' - a_{23}' x_3)}{a_{22}'}; x_1 = \frac{(c_1' - a_{12}' x_2 - a_{13}' x_3)}{a_{11}}$$

Gauss Elimination method will fail immediately because the element in the first row on the leading diagonal, the pivot is zero. Thus, it is impossible to divide that row by the pivot value.

3.7.1 Stepwise Solution to Solve the Problems by Gauss Elimination Method (Illustrated by Numerical Example)

Example 3.7.1

Solve the following equations using Gauss Elimination method.

$$2x_1 + 3x_2 + 4x_3 = 11$$

$$9x_1 + 2x_2 - 8x_3 = 1.9$$

$$15x_1 - 8x_2 + 6x_3 = 14.7$$

Solution :

Step 1 : Convert the given equation into matrix form $AX = B$

$$\therefore \begin{bmatrix} 2 & 3 & 4 \\ 9 & 2 & -8 \\ 15 & -8 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 11 \\ 1.9 \\ 14.7 \end{bmatrix}$$

Step 2 : Convert 1st diagonal element as 1

Note : By converting 1st diagonal element as 1, we can convert remaining elements under that diagonal element (of the same column) zero very easily.

$$\therefore R_1 \Rightarrow (R_1) \times (1/2)$$

$$\begin{bmatrix} 1 & 1.5 & 2 \\ 9 & 2 & -8 \\ 15 & -8 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5.5 \\ 1.9 \\ 14.7 \end{bmatrix}$$

Step 3 : Convert remaining elements (of 1st column) under 1 as zero

$$R_2 \Rightarrow R_2 - 9R_1, R_3 \Rightarrow R_3 - 15R_1$$

$$\begin{bmatrix} 1 & 1.5 & 2 \\ 0 & -11.5 & -26 \\ 0 & -30.5 & -24 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5.5 \\ -47.59 \\ -67.8 \end{bmatrix}$$

Step 4 : Convert 2nd diagonal element as 1

Note : By converting 2nd diagonal element as 1, we can convert remaining elements under that diagonal element (of the same column) zero very easily.

$$\therefore R_2 \Rightarrow R_2 \times \frac{1}{-11.5}$$

$$\therefore \begin{bmatrix} 1 & 1.5 & 2 \\ 0 & 1 & 2.26 \\ 0 & -30.5 & -24 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5.5 \\ 4.139 \\ -67.8 \end{bmatrix}$$

Step 5 : Convert remaining elements (of 2nd column) under 1 as zero

$$R_3 \Rightarrow R_3 + 30.5R_2$$

$$\begin{bmatrix} 1 & 1.5 & 2 \\ 0 & 1 & 2.26 \\ 0 & 0 & 44.95 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5.5 \\ 4.139 \\ 58.44 \end{bmatrix}$$

Step 6 : Convert 3rd diagonal element as 1

$$R_3 \Rightarrow R_3 (1/44.95)$$

$$\begin{bmatrix} 1 & 1.5 & 2 \\ 0 & 1 & 2.26 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5.5 \\ 4.139 \\ 1.3 \end{bmatrix}$$

There are no future elements available under this diagonal element (for 3rd column), hence we have to apply backward substitution for the solution.

Step 7 : Apply backward substitution we get,

$$x_3 = 1.3$$

and $x_2 + 2.26x_3 = 4.139$

$$\therefore x_2 = 4.139 - 2.26(1.3) = 1.2 \quad (\because x_3 = 1.3)$$

and $x_1 + 1.5x_2 - 2x_3 = 5.5$

$$\therefore x_1 = 5.5 - 1.5(1.2) + 2(1.3) \quad (\because x_2 = 1.2 \text{ and } x_3 = 1.3)$$

$$\therefore x_1 = 1.1$$

$$x_1 = 1.1, x_2 = 1.2, x_3 = 1.3$$

...Ans.





Note : Flowchart and Program on Gauss Elimination method are explained in Chapter 9.

Example 3.7.2

Using Gauss Elimination Method, solve the following set of simultaneous equations.

$$2x + 4y - 6z = -4;$$

$$x + 5y + 3z = 10;$$

$$x + 3y + 2z = 5$$

Solution :

Step 1 : Write the equation in matrix form.

$$\begin{bmatrix} 2 & 4 & -6 \\ 1 & 5 & 3 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -4 \\ 10 \\ 5 \end{bmatrix}$$

Step 2 : Making 1st element in 1st column as 1.

$$R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 5 & 3 \\ 2 & 4 & -6 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ -4 \\ 5 \end{bmatrix}$$

Step 3 : Making other elements in column 1 as 0

$$R_2 = R_2 - 2R_1, R_3 = R_3 - R_1$$

$$\begin{bmatrix} 1 & 5 & 3 \\ 0 & -6 & -12 \\ 0 & -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ -24 \\ -5 \end{bmatrix}$$

Step 4 : Making element in 2nd column as 1.

$$R_2 = R_2 / (-6)$$

$$\begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 2 \\ 0 & -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \\ -5 \end{bmatrix}$$

Step 5 : Making element in 3rd row and 2nd column as 0.

$$R_3 = R_3 + 2R_2$$

$$\begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \\ 3 \end{bmatrix}$$

Step 6 : Making element in 3rd column and 3rd row as 1

$$R_3 = \frac{R_3}{3}$$

$$\begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \\ 1 \end{bmatrix}$$

Step 7 : Converting into equation form and solving by back substitution.

$$z = 1;$$

$$y + 2z = 4;$$

$$y + 2 = 4$$

$$\therefore y = 2;$$

$$x + 5y + 3z = 10$$

$$x + 5(2) + 3(1) = 10$$

$$\therefore x = -3$$

Solution of equation is

$$x = -3, y = 2, z = 1$$

...Ans.

Example 3.7.3

Apply Gauss-Elimination method to solve the equations as follows :

$$x + 3y + 3z = 16$$

$$x + 4y + 3z = 18$$

$$x + 3y + 4z = 19$$

Solution :

Given data :

$$x + 3y + 3z = 16$$

$$x + 4y + 3z = 18$$

$$x + 3y + 4z = 19$$

Using Gauss elimination find $x = ?$, $y = ?$, $z = ?$

Step 1 : Convert the given equation into matrix form $AX = B$

$$\therefore \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ 18 \\ 19 \end{bmatrix}$$

Step 2 : Convert 1st diagonal element as 1 easily : already 1

Step 3 : Convert remaining elements (of 1st column) under 1 as zero

$$R_2 \Rightarrow R_2 - R_1, R_3 \Rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ 2 \\ 3 \end{bmatrix}$$

Step 4 & 5 : Diagonal element as 1. Convert remaining elements (of 2nd column) under a_{22} as zero

Step 6 : Apply backward substitution we get,

$$\therefore z = 3, y = 2$$

$$\text{and } x + 3y + 3z = 16$$

$$x = 16 - 6 - 9 = 1$$

$$x = 1, y = 2, z = 3$$

...Ans.

Example 3.7.4

Find the numerical solution of the system of equations $x - y + 4z = 16$, $3x + 2y + z = 18$ and $x + 4y - 2z = 12$, using Gauss elimination method.

Solution :

Given data :

$$x - y + 4z = 16; 3x + 2y + z = 18; x + 4y - 2z = 12$$

Using Gauss elimination find $x = ?$, $y = ?$, $z = ?$

Step 1 : Convert the given equation into matrix form $AX = B$

$$\therefore \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & 1 \\ 1 & 4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ 18 \\ 12 \end{bmatrix}$$

Step 2 : Convert 1st diagonal element as 1 easily : already 1

Step 3 : Convert remaining elements (of 1st column) under 1 as zero

$$R_2 \Rightarrow R_2 - 3R_1, R_3 \Rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & -1 & 4 \\ 0 & 5 & -11 \\ 0 & 5 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ -30 \\ -4 \end{bmatrix}$$



Step 4: Convert 2nd diagonal element as 1 easily.

$$\therefore R_2 \Rightarrow R_2 \times \frac{1}{5} \text{ and } R_3 \Rightarrow R_3 \times \frac{1}{5}$$

$$\begin{bmatrix} 1 & -1 & 4 \\ 0 & 1 & -11/5 \\ 0 & 1 & -6/5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ -6 \\ -4/5 \end{bmatrix}$$

Step 5: Convert remaining elements (of 2nd column) under 1 as zero

$$\therefore R_3 \Rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & -1 & 4 \\ 0 & 1 & -11/5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ -6 \\ 26/5 \end{bmatrix}$$

Step 6: Convert 3rd diagonal element as 1: already 1

Step 7: Apply backward substitution we get,

$$\therefore z = 26/5 = 5.2$$

$$\text{and } y - z(11/5) = -6$$

$$\therefore y = 136/25 = 5.44$$

$$\text{and } x - y + 4z = 16$$

$$x = 16 - (136/25) - 4(26/5)$$

$$x = 16/25 = 0.64$$

$$\therefore x = 0.64, y = 5.44, z = 5.2$$

...Ans.

Example 3.7.5

Solve the following set of simultaneous equations using Gauss Elimination Method.

$$10x_1 + x_2 + x_3 = 12$$

$$2x_1 + 11x_2 + 2x_3 = 15$$

$$3x_1 + 4x_2 + 9x_3 = 16$$

Solution:

$$10x_1 + x_2 + x_3 = 12$$

$$2x_1 + 11x_2 + 2x_3 = 15$$

$$3x_1 + 4x_2 + 9x_3 = 16$$

Step 1: Convert the given equation in matrix form $AX = B$

$$\begin{bmatrix} 10 & 1 & 1 \\ 2 & 11 & 2 \\ 3 & 4 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 15 \\ 16 \end{bmatrix}$$

Step 2: Convert 1st diagonal element as 1.

$$R_1 \Rightarrow \frac{1}{10}(R_1)$$

$$\begin{bmatrix} 1 & 0.1 & 0.1 \\ 2 & 11 & 2 \\ 3 & 4 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 15 \\ 16 \end{bmatrix}$$

Step 3: Convert remaining elements (of 1st column) under 1 as 0.

$$R_2 = R_2 - 2R_1, R_3 = R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 0.1 & 0.1 \\ 0 & 10.8 & 1.8 \\ 0 & 3.7 & 8.7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 12.6 \\ 12.4 \end{bmatrix}$$

Step 4: Convert second diagonal element as 1.

$$R_2 \Rightarrow \frac{1}{10.8} R_2, R_3 \Rightarrow \frac{1}{3.7} R_3$$

$$\begin{bmatrix} 1 & 0.1 & 0.1 \\ 0 & 1 & 0.1667 \\ 0 & 1 & 2.351 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 1.166 \\ 3.351 \end{bmatrix}$$

Step 5: Convert remaining element (of 2nd column) under 1 as zero.

$$R_3 - R_2$$

$$\begin{bmatrix} 1 & 0.1 & 0.1 \\ 0 & 1 & 0.1667 \\ 0 & 0 & 2.1843 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 1.166 \\ 2.185 \end{bmatrix}$$

Step 6: Convert third diagonal element as 1.

$$R_3 \Rightarrow \frac{1}{2.1843} R_3$$

$$\begin{bmatrix} 1 & 0.1 & 0.1 \\ 0 & 1 & 0.1667 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 1.166 \\ 1 \end{bmatrix}$$

Step 7: Apply backward substitution

$$z = 1$$

$$y + 0.1667z = 1.166$$

$$\therefore y = 1$$

$$x + 0.1y + 0.1z = 1.2$$

$$x = 1.2 - 0.1 - 0.1 = 1$$

$$\therefore x = 1, y = 1, z = 1$$

...Ans.

Example 3.7.6

Solve the following system of equation using Gauss elimination method.

$$3x + 2y + 3z = 18;$$

$$2x + y + z = 10;$$

$$x + 4y + 9z = 16$$

Solution:

Given Data:

$$3x + 2y + 3z = 18;$$

$$2x + y + z = 10;$$

$$x + 4y + 9z = 16$$

Step 1: Convert the equation into matrix form $AX = B$

$$\begin{bmatrix} 3 & 2 & 3 \\ 2 & 1 & 1 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 18 \\ 10 \\ 16 \end{bmatrix}$$

Step 2: Convert 1st diagonal element as 1.

$$\therefore R_1 \Rightarrow R_1 \times \left(\frac{1}{3}\right)$$

$$\begin{bmatrix} 1 & 0.6666 & 1 \\ 2 & 1 & 1 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 16 \end{bmatrix}$$

Step 3: Convert remaining elements (of 1st column) order 1 as zero.

$$\therefore R_2 \Rightarrow R_2 - 2R_1, R_3 \Rightarrow R_3 - R_1$$



$$\begin{bmatrix} 1 & 0.6666 & 1 \\ 0 & -0.3332 & -1 \\ 0 & 3.3334 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \\ 10 \end{bmatrix}$$

Step 4: Convert 2nd diagonal element as 1.

$$R_2 = R_1 \times \frac{1}{(-0.3332)}$$

$$\begin{bmatrix} 1 & 0.6666 & 1 \\ 0 & 1 & 3 \\ 0 & 3.3334 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ 10 \end{bmatrix}$$

Step 5: Convert remaining element (of 2nd column) under 1 as:

$$R_3 \Rightarrow R_3 - 3.3334 R_2$$

$$\begin{bmatrix} 1 & 0.6666 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ -10 \end{bmatrix}$$

Step 6: Convert 3rd diagonal element as:

$$R_3 \Rightarrow R_3 \times \left(\frac{1}{-2}\right)$$

$$\begin{bmatrix} 1 & 0.6666 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ 5 \end{bmatrix}$$

Arrange matrix property

Step 7: Apply backward substitution, we get

$$\begin{aligned} z &= 5 \\ y + 3z &= 6 \\ \therefore y &= 6 - 3z \\ \therefore y &= -9 \\ x + 0.6666y + z &= 6 \\ \therefore x &= 6 - 0.6666(-9) - 5 \\ \therefore x &= 7 \\ \therefore x &= 7, y = -9, z = 5 \quad \dots \text{Ans.} \end{aligned}$$

Ex. 3.7.7

Solve the following set of linear simultaneous equation using Gauss elimination method.

$$x + 3y + z = 10$$

$$x + 2y + 5z = 12$$

$$4x + y + 2z = 16$$

Solution:

Given Equations:

$$x + 3y + z = 10$$

$$x + 2y + 5z = 12$$

$$4x + y + 2z = 16$$

Step 1: Convert equation into matrix form $AX = B$

$$\begin{bmatrix} 1 & 3 & 1 \\ 1 & 2 & 5 \\ 4 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 12 \\ 16 \end{bmatrix}$$

Step 2: Convert remaining elements (of 1st column) order 1 as zero.

$$R_2 \Rightarrow R_2 - R_1, R_3 \Rightarrow R_3 - 4R_1$$

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & -4 \\ 0 & -11 & -2 \end{bmatrix} X = \begin{bmatrix} 10 \\ -2 \\ -24 \end{bmatrix}$$

Step 3: Convert remaining element of 2nd column order 1 as zero

$$R_3 \Rightarrow R_3 + 11R_2$$

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & -4 \\ 0 & 0 & -46 \end{bmatrix} X = \begin{bmatrix} 10 \\ -2 \\ -46 \end{bmatrix}$$

Step 4: Apply back substitution we get,

$$-46z = -46 \quad \therefore z = 1$$

$$y - 4z = -2 \quad \therefore y = 2$$

$$x + 3y + z = 10$$

$$x + 6 + 1 = 10 \quad \therefore x = 3$$

$$\therefore \text{Solution is } x = 3, y = 2, z = 1$$

Ex. 3.7.8

Using Gauss elimination method solve following set of simultaneous equations

$$x_1 + 3x_2 + 5x_3 = 2$$

$$3x_1 + 2x_2 + 4x_3 = 7$$

$$2x_1 + x_2 + x_3 = 4$$

Solution:

Step 1: Convert equation into matrix form $AX = B$

$$\begin{bmatrix} 1 & 3 & 5 \\ 3 & 2 & 4 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 4 \end{bmatrix}$$

Step 2: Convert remaining elements (of 1st column) order 1 as zero.

$$R_2 = R_2 - 3R_1; R_3 = R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 3 & 5 \\ 0 & -7 & -11 \\ 0 & -5 & -9 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Step 3: Convert 2nd diagonal element as 1

$$R_2 \Rightarrow R_2 \times \left(\frac{-1}{7}\right)$$

$$\begin{bmatrix} 1 & 3 & 5 \\ 0 & 1 & 1.5714 \\ 0 & -5 & -9 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -0.1428 \\ 0 \end{bmatrix}$$

Step 4: Convert remaining elements of 2nd column order 1 as zero

$$R_3 = R_3 + 5R_2$$

$$\begin{bmatrix} 1 & 3 & 5 \\ 0 & 1 & 1.5714 \\ 0 & 0 & -1.143 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -0.1428 \\ -0.714 \end{bmatrix}$$

Step 5: Convert 3rd diagonal element as 1

$$R_3 = R_3 \times \left(\frac{-1}{1.143}\right)$$

$$\begin{bmatrix} 1 & 3 & 5 \\ 0 & 1 & 1.5714 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -0.1428 \\ 0.6246 \end{bmatrix}$$

Step 6 : Apply backward substitution we get

$$x_3 = 0.625$$

$$\text{and } x_2 + 1.5714 x_3 = -0.1428$$

$$\begin{aligned} x_2 &= -0.428 - 1.5714 x_3 \\ &= -0.428 - (1.5714 \times 0.625) \end{aligned}$$

$$x_2 = -1.125$$

$$\text{and } x_1 + 3x_2 + 5x_3 = 2$$

$$x_1 = 2 - 3x_2 - 5x_3 = 2 - (3 \times -1.125) - (5 \times 0.625)$$

$$x_1 = 2.235$$

$$x_1 = 2.235, x_2 = -1.12, x_3 = 0.625$$

3.8 Drawbacks of Elimination Methods : (Pitfall Technique)

3.8.1 Division by Zero

Review Question

Q. Write a short note on pitfalls of Gauss elimination method.

There is quite a possibility of division by zero where the first equation may have coefficient to be zero or the second equation having second coefficient to be zero and so on. For example,

$$2x_2 + 3x_3 = 8 \quad \text{or} \quad 0x_1 + 2x_2 + 3x_3 = 8$$

$$9x_1 + 5x_2 + 7x_3 = -3$$

$$5x_1 - 2x_2 + 6x_3 = 5$$

Here during backward substitution since coefficient of x_1 is zero, division by zero will create a big problem. The technique of pivoting has been developed to partially avoid these problems.

Syllabus Topic : Pivoting

3.8.2 Pivoting

Review Questions

Q. What is meant by pivoting in Gauss elimination to solve simultaneous equations?

Q. Write a short note on pivoting.

- As discussed, there might be a problem if the pivot element is zero.
- It induces division by zero during backward substitution. Also round off errors are introduced if the pivot element is very small as compared to other elements.
- It is therefore, always better to determine largest available coefficient before normalizing (converting to upper triangular form).
- The rows can then be switched so that the largest element is the pivot element. This is called partial pivoting. If columns as well as rows are searched for the largest element and then switched, the procedure is called complete pivoting.

Complete pivoting is rarely used because switching columns changes the order of the x 's and consequently adds significant and usually unjustified complexity to the computer program.

Pivoting avoids division by zero as well as reduces round off errors. As such, it also serves a partial remedy for ill conditioning.

Procedure for Solving Equations using Gauss Elimination Method with Partial Pivoting

Consider the equation

$$a_1x + b_1y + c_1z = P$$

$$a_2x + b_2y + c_2z = Q$$

$$a_3x + b_3y + c_3z = R$$

Such that $a_2 > a_1$ and a_3

Note : The coefficient can be compared only according to their magnitude.

Step 1 : Convert the equation into matrix form $AX = B$

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} P \\ Q \\ R \end{bmatrix}$$

Step 2 : Find the maximum element in the first column. ($\because a_2 > a_1$ and a_3) then replace the row existing that element with 1st row.

$\therefore$ Replacing R_1 by R_2

$$\therefore \begin{bmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} Q \\ P \\ R \end{bmatrix}$$

Step 3 : Eliminate x from second and third row

$\therefore$ It is possible by normal gauss elimination method, explained earlier

Then the coefficient matrix appear as

$$\begin{bmatrix} a_2 & b_2 & c_2 \\ 0 & b'_1 & c'_1 \\ 0 & b'_3 & c'_3 \end{bmatrix}$$

Step 4 : Now from 2nd and 3rd row find the largest element in the second column. Then replace the row existing that largest element with 2nd row. If $b'_3 > b'_1$ then

The coefficient matrix appear as

$$\begin{bmatrix} a_2 & b_2 & c_2 \\ 0 & b'_3 & c'_3 \\ 0 & b'_1 & c'_1 \end{bmatrix}$$

Step 5 : Eliminate y from third row. It is possible by again using normal gauss elimination method explain earlier.

Then the coefficient of matrix appear as

$$\begin{bmatrix} a_2 & b_2 & c_2 \\ 0 & b'_3 & c'_3 \\ 0 & 0 & c''_3 \end{bmatrix}$$

Step 6: Apply backward substitution method to obtain the solution i.e. (the value of x , y , and z)

3.8.3 Stepwise Solution to Solve the Problems by Gauss Elimination Method With Partial Pivoting (Illustrated by Numerical Example)

Example 3.8.1

Using Gauss elimination method with partial pivoting, solve the following equations.

$$5x + 13y + 7z = 14$$

$$3x + 6y + 3z = 9$$

$$7x + 2y + 4z = 5$$

Solution :

Given data : $5x + 13y + 7z = 14$, $3x + 6y + 3z = 9$,
 $7x + 2y + 4z = 5$

Step 1 : Convert the equation into matrix form $AX = B$

$$\begin{bmatrix} 5 & 13 & 7 \\ 3 & 6 & 3 \\ 7 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 14 \\ 9 \\ 5 \end{bmatrix}$$

Step 2 : Find the maximum element in the first column.

Maximum element in 1st column is 7 and it is present in third row.

$$\therefore R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} 7 & 2 & 4 \\ 3 & 6 & 3 \\ 5 & 13 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \\ 14 \end{bmatrix}$$

Step 3 : Convert 1st diagonal element as 1. $R_1 \Rightarrow R_1 \times \left(\frac{1}{7}\right)$

$$\begin{bmatrix} 1 & 0.285 & 0.571 \\ 3 & 6 & 3 \\ 5 & 13 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0.7142 \\ 9 \\ 14 \end{bmatrix}$$

Step 4 : By normal Gauss elimination method convert remaining element (of 1st column) under 1 as zero

$$R_2 \Rightarrow R_2 - 3R_1, \quad R_3 \Rightarrow R_3 - 5R_1$$

$$\begin{bmatrix} 1 & 0.285 & 0.571 \\ 0 & 5.142 & 1.285 \\ 0 & 11.575 & 4.145 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0.7142 \\ 6.8571 \\ 10.429 \end{bmatrix}$$

Step 5 : Now, maximum element in the second column from 2nd and 3rd row is 11.575 and it is present in the third row

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 0.285 & 0.571 \\ 0 & 11.575 & 4.145 \\ 0 & 5.142 & 1.285 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0.7142 \\ 10.429 \\ 6.8571 \end{bmatrix}$$

Step 6 : By normal Gauss elimination method convert 2nd diagonal element as 1

$$R_2 = R_2 \times \frac{1}{11.575}$$

$$\begin{bmatrix} 1 & 0.285 & 0.571 \\ 0 & 1 & 0.3579 \\ 0 & 5.142 & 1.285 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0.7142 \\ 0.9012 \\ 6.8571 \end{bmatrix}$$

Step 7 : Converting remaining element (of 2nd column) under 1 as zero

$$\begin{bmatrix} 1 & 0.285 & 0.571 \\ 0 & 1 & 0.3579 \\ 0 & 0 & -0.5553 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0.7142 \\ 0.9012 \\ 2.2231 \end{bmatrix}$$

Step 8 : Convert 3rd diagonal element as 1

$$\begin{bmatrix} 1 & 0.285 & 0.571 \\ 0 & 1 & 0.3579 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0.7142 \\ 0.9012 \\ -4.005 \end{bmatrix}$$

Step 9 : Apply backward substitution method

$$\therefore z = -4.005$$

$$\text{and } y + 0.3572 = 0.9012$$

$$\therefore y = 0.9012 - 0.357(-4.005) = 2.33$$

$$\text{and } x + 0.285y + 0.571z = 0.7142$$

$$\therefore x = 0.7142 - 0.285(2.33)$$

$$+ 0.571 \times (-4.005)$$

$$\therefore x = 2.332$$

$$\therefore x = 2.332, \quad y = 2.33, \quad z = -4.005 \quad \dots \text{Ans.}$$

Example 3.8.2

Use Gauss Elimination Method to solve the following equation. Use partial pivoting.

$$x_1 + 20x_2 + x_3 = 22$$

$$-x_1 - x_2 + 20x_3 = 18$$

$$20x_1 + x_2 - x_3 = 20$$

Solution :

$$\text{Given data : } x_1 + 20x_2 + x_3 = 22$$

$$-x_1 - x_2 + 20x_3 = 18$$

$$20x_1 + x_2 - x_3 = 20$$

Step 1 : Convert the equation into matrix form $AX = B$

$$\begin{bmatrix} 1 & 20 & 1 \\ -1 & -1 & 20 \\ 20 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 22 \\ 18 \\ 20 \end{bmatrix}$$

Step 2 : Find the maximum element in the first column.

Maximum element 20 is present in 3rd row

$$\therefore R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} 20 & 1 & -1 \\ -1 & -1 & 20 \\ 1 & 20 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 18 \\ 22 \end{bmatrix}$$

Step 3 : Convert the 1st diagonal element as 1

$$R_1 \Rightarrow R_1 \times \left(\frac{1}{20}\right)$$

$$\begin{bmatrix} 1 & 0.05 & -0.05 \\ -1 & -1 & 20 \\ 1 & 20 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 18 \\ 22 \end{bmatrix}$$

Step 4 : Convert remaining element under 1 as zero

$$R_2 \Rightarrow R_2 + R_1$$

$$R_3 = R_3 - R_1$$



$$\begin{bmatrix} 1 & 0.05 & -0.05 \\ 0 & -0.95 & 19.95 \\ 0 & 19.95 & 1.05 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 19 \\ 21 \end{bmatrix}$$

Step 5: Maximum element in second column from 2nd and 3rd row is 19.95 and it is present in 3rd row.

$$R_2 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & 0.05 & -0.05 \\ 0 & 19.95 & 1.05 \\ 0 & -0.95 & 19.95 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 21 \\ 19 \end{bmatrix}$$

Step 6: Convert 2nd diagonal element as 1

$$R_2 \Rightarrow R_2 \times \frac{1}{19.95}$$

$$\begin{bmatrix} 1 & 0.05 & -0.05 \\ 0 & 1 & 0.05 \\ 0 & -0.95 & 19.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1.05 \\ 19 \end{bmatrix}$$

Step 7: Convert remaining element in 2nd column under 1 as zero

$$R_3 = R_3 + (0.95 R_2)$$

$$\begin{bmatrix} 1 & 0.05 & -0.05 \\ 0 & 1 & 0.05 \\ 0 & 0 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1.05 \\ 20 \end{bmatrix}$$

Step 8: Convert 3rd diagonal element as 1

$$R_3 \Rightarrow R_3 \times \left(\frac{1}{20}\right)$$

$$\begin{bmatrix} 1 & 0.05 & -0.05 \\ 0 & 1 & 0.05 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1.05 \\ 1 \end{bmatrix}$$

Step 9: Apply backward substitution so, $x_3 = 1$

$$1(x_2) + 0.05(x_3) = 1.05$$

$$x_2 + 0.05(1) = 1.05$$

$$x_2 = 1.05 - 0.05$$

$$x_2 = 1$$

$$\text{and } x_1 + 0.05(x_2) - 0.05(x_3) = 1$$

$$x_1 + 0.05(1) - 0.05(1) = 1$$

$$x_1 = 1$$

$$\therefore x_1 = 1, x_2 = 1, x_3 = 1$$

...Ans.

Example 3.8.3

Solve the system of equation using Gauss elimination method

$$x + y - 2z = 3$$

$$4x - 2y + z = 5$$

$$3x - y + 3z = 8$$

If required use partial pivoting.

Solution:

Step 1: Convert given equation into matrix form $AX = B$

$$\begin{bmatrix} 1 & 1 & -2 \\ 4 & -2 & 1 \\ 3 & -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 8 \end{bmatrix}$$

Step 2: Find maximum element in the first column
maximum element is 4 and it is present in 2nd column.

$$\therefore R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 4 & -2 & 1 \\ 1 & 1 & -2 \\ 3 & -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ 8 \end{bmatrix}$$

Step 3: Convert the 1st diagonal element as 1

$$R_1 = \frac{R_1}{4}$$

$$\begin{bmatrix} 1 & -0.5 & 0.25 \\ 1 & 1 & -2 \\ 3 & -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1.25 \\ 3 \\ 8 \end{bmatrix}$$

Step 4: Convert remaining element under 1 as zero

$$R_2 = R_2 - R_1 \text{ and } R_3 = R_3 - 3R_1$$

$$\begin{bmatrix} 1 & -0.5 & 0.25 \\ 0 & 1.5 & -2.25 \\ 0 & 0.5 & 2.25 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1.25 \\ 1.75 \\ 4.25 \end{bmatrix}$$

Step 5: Now maximum element in the second column from 2nd and 3rd row is 1.5 and it is already present in the 2nd row so proceed to next step.

Step 6: Convert 2nd diagonal element as 1

$$R_2 = R_2 \times \frac{1}{1.5}$$

$$\begin{bmatrix} 1 & -0.5 & 0.25 \\ 0 & 1 & -1.5 \\ 0 & 0.5 & 2.25 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1.25 \\ 1.666 \\ 4.25 \end{bmatrix}$$

Step 7: Convert remaining element under 1 as zero.

$$R_3 = R_3 - (0.5 R_2)$$

$$\begin{bmatrix} 1 & -0.5 & 0.25 \\ 0 & 1 & -1.5 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1.25 \\ 1.666 \\ 3.417 \end{bmatrix}$$

Step 8: Convert 3rd diagonal element as 1

$$R_3 = R_3 \times \frac{1}{3}$$

$$\begin{bmatrix} 1 & -0.5 & 0.25 \\ 0 & 1 & -1.5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1.25 \\ 1.666 \\ 1.139 \end{bmatrix}$$

Step 9: Apply backward substitution

$$\text{So, } z = 1.222$$

$$y - 1.5z = 1.666$$

$$\therefore y = 3$$

$$\text{and } x - 0.5y + 0.25z = 1.25$$

$$\therefore x = 2.444$$

$$x = 2.444, y = 3, z = 1.222$$

Syllabus Topic : Engineering Applications

3.9 Engineering Applications

Linear algebraic system of equations occurs in many applications. The simplest case is two equations with two unknown. But there are few cases where a system is with ten, fifty or even hundred equations and unknown.

1. In the Fig. 3.9.1 there is an arrangement of 4 springs connected in series being compressed by a force F .



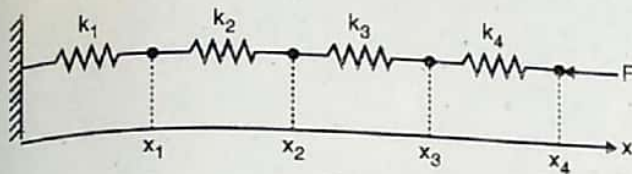


Fig. 3.9.1

At equilibrium force balance equations can be developed defining the relationship between the spring.

$$k_2(x_2 - x_1) = k_1 x_1$$

$$k_3(x_3 - x_2) = k_2(x_2 - x_1)$$

$$k_4(x_4 - x_3) = k_3(x_3 - x_2)$$

$$F = k_4(x_4 - x_3)$$

where 'k' are spring constants. If a values of k_1 , k_2 , k_3 and k_4 are known, then we can calculate values of x_1 , x_2 , x_3 and x_4 a by solving simultaneous equations in form of x.

2. For a given ladder, the circuit can be given as shown in Fig. 3.9.2.

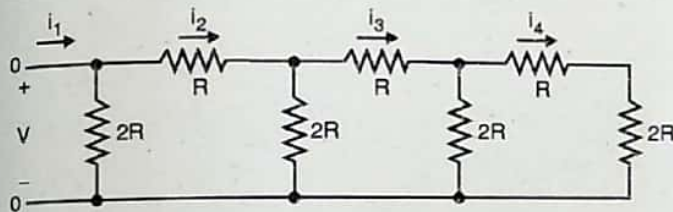


Fig. 3.9.2

Applying Kirchoff's voltage law to each mesh of ladder and diving through by the Resistance R, then we can find current i_1 , i_2 , i_3 and i_4 by solving simultaneous equations in terms of i.

$$\therefore 2i_1 - 2i_2 = \frac{V}{R}$$

$$-2i_1 + 5i_2 - 2i_3 = 0$$

$$-2i_2 + 5i_3 - 2i_4 = 0$$

3.9.1 Engineering Applications Problems

Example 3.9.1

Current in a circuit are given by the following equations :

$$28 I_1 - 3 I_2 = 10,$$

$$-3 I_1 + 38 I_2 - 5 I_5 = 0,$$

$$-10 I_2 + 25 I_3 - 15 I_4 = 0,$$

$$-15 I_3 + 45 I_4 = 0,$$

$$-5 I_2 + 30 I_5 = 0$$

Estimate the current using Gauss elimination method.

Solution :

Given data : The given equations are :

$$28 I_1 - 3 I_2 = 10$$

$$-3 I_1 + 38 I_2 - 5 I_5 = 0$$

$$-10 I_2 + 25 I_3 - 15 I_4 = 0$$

$$-15 I_3 + 45 I_4 = 0$$

$$-5 I_2 + 30 I_5 = 0$$

Step 1 : Convert the given equation into matrix form $AX = B$

$$\begin{bmatrix} 28 & -3 & 0 & 0 & 0 \\ -3 & 38 & 0 & 0 & -5 \\ 0 & -10 & 25 & -15 & 0 \\ 0 & 0 & -15 & 45 & 0 \\ 0 & -5 & 0 & 0 & 30 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 2 : Convert 1st diagonal element as 1

Note : By converting 1st diagonal element as 1, we can convert remaining elements under that diagonal element (of the same column) zero very easily.

$$R_1 \Rightarrow R_1 \times \frac{1}{28}$$

$$\begin{bmatrix} 1 & -0.107 & 0 & 0 & 0 \\ -3 & 38 & 0 & 0 & -5 \\ 0 & -10 & 25 & -15 & 0 \\ 0 & 0 & -15 & 45 & 0 \\ 0 & -5 & 0 & 0 & 30 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 0.357 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 3 : Convert remaining element of 1st column under 1 as zero

$$\therefore R_2 \Rightarrow R_2 + 3R_1$$

$$\begin{bmatrix} 1 & -0.107 & 0 & 0 & 0 \\ 0 & 37.67 & 0 & 0 & -5 \\ 0 & -10 & 25 & -15 & 0 \\ 0 & 0 & -15 & 45 & 0 \\ 0 & -5 & 0 & 0 & 30 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 0.357 \\ 1.071 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 4 : Convert 2nd diagonal element as 1

$$R_2 \Rightarrow R_2 \times \frac{1}{37.67}$$

$$\begin{bmatrix} 1 & -0.107 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -0.132 \\ 0 & -10 & 25 & -15 & 0 \\ 0 & 0 & -15 & 45 & 0 \\ 0 & -5 & 0 & 0 & 30 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 0.357 \\ 0.0285 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 5 : Convert all the element of the 2nd column under diagonal element 1 as zero.

$$R_3 \Rightarrow R_3 + 10R_2, R_5 \Rightarrow R_5 + 5R_2$$

$$\begin{bmatrix} 1 & -0.107 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -0.132 \\ 0 & 0 & 25 & -15 & -1.327 \\ 0 & 0 & 25 & -15 & -1.327 \\ 0 & 0 & -15 & 45 & 0 \\ 0 & 0 & 0 & 0 & 29.33 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 0.357 \\ 0.0284 \\ 0.28436 \\ 0 \\ 0.1421 \end{bmatrix}$$

Sep 6 : Convert 3rd diagonal element as 1

$$\therefore R_3 \Rightarrow R_3 \times \frac{1}{25}$$

Step 7 : Convert remaining element of 3rd column under 1 as zero.

$$\therefore R_4 \Rightarrow R_4 + 15R_3$$

Step 8: Convert 4th diagonal element as 1

$$\therefore R_4 \Rightarrow R_4 \times \frac{1}{36}$$

As element under diagonal element of 4th column is already zero so go for next step.

Step 9: Convert 5th diagonal element as 1

$$R_5 \Rightarrow R_5 \times \frac{1}{29.33}$$

After performing steps 6 to 9 the matrix will appear in the form as :

$$\begin{bmatrix} 1 & -0.107 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -0.132 \\ 0 & 0 & 1 & -0.6 & -0.05 \\ 0 & 0 & 0 & 1 & -0.022 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 0.357 \\ 0.0284 \\ 0.01137 \\ 0.0047 \\ 0.004847 \end{bmatrix}$$

Step 10: Apply backward substitution we get

$$I_5 = 0.004847 \quad I_4 = 0.004847 \quad I_3 = 0.0145$$

$$I_2 = 0.02907, \quad I_1 = 0.36025$$

...Ans.

Example 3.9.2

Use Gauss elimination to find i_1, i_2, i_3 . What are the pitfalls in G-E method? (all resistances are in ohms).

Refer Fig. P. 3.9.2

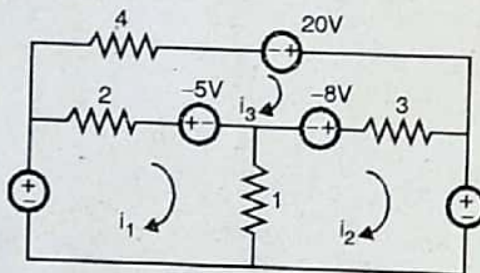


Fig. P. 3.9.2

Solution :

Assumption : Here values of two voltage sources are given we assume them to be 10 V each

Writing equation of i_1, i_2, i_3 as,

$$4(i_3) + 3(i_3 - i_2) + 2(i_3 - i_1) = 20 - 8 + 5$$

$$2(i_1 - i_3) + 1(i_1 - i_2) = -5 + 10$$

$$3(i_2 - i_3) + 1(i_2 - i_1) = +8 - 10$$

$$\text{Simplifying } -2i_1 - 6i_2 + 9i_3 = 17$$

$$3i_1 - i_2 - 2i_3 = 5$$

$$-i_1 - 4i_2 - 3i_3 = -2$$

Step 1: Rearranging equation so that diagonal elements are largest

$$3i_1 - i_2 - 2i_3 = 5$$

$$-i_1 - 4i_2 - 3i_3 = -2$$

$$-2i_1 - 6i_2 + 9i_3 = 17$$

Step 2: Convert the matrix into $AI = B$ form

$$\begin{bmatrix} 3 & -1 & -2 \\ -1 & 4 & -3 \\ -2 & -6 & 9 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 17 \end{bmatrix}$$

Step 3: Convert the matrix in the Gauss Elimination format.

$$\begin{bmatrix} 1 & -0.33 & -0.666 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1.66 \\ -0.0909 \\ 19.72 \end{bmatrix}$$

Step 4: Apply backward substitution method

$$\therefore i_3 = 19.7274, i_2 = 19.6363, i_1 = 21.2712,$$

$$i_1 = 20.3636, i_2 = 19.6363, i_3 = 19.7274 \quad \dots \text{Ans.}$$

...Chapter End

□□□

CHAPTER 4 UNIT IV

Numerical Integration and Differentiation

Syllabus :

Motivation, Newton's Cotes Integration Formulas : Trapezoidal Rule, Simpson's rule, Engineering applications, Numerical differentiation using Finite divide Difference method

Syllabus Topic : Motivation

4.1 Introduction

- Engineering applications always deals with system and processes that changes with time. Engineer's duty is to draw various conclusions about the system from an analysis of its mathematical model.
- To 'Integrate' word suggests to bring close i.e. to unite or to combine the parts as a whole. One can say integration is to compute the area enclosed by a curve, but an Engineer can say integration is the process of calculating integral from a set of tabulated values of specific function.

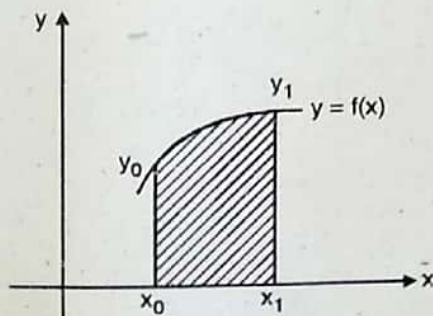


Fig. 4.1.1

- As shown in Fig. 4.1.1, curve is a plot of a function $y = f(x)$. The integration of the curve or function is the area under the curve.

$$\therefore \text{The shaded area} = A = \int_{x_0}^{x_1} f(x) \cdot dx$$

- The above equation is the definite integral of $f(x)$ between lower limit x_0 and upper limit x_1 . The equation is also called quadrature formulae.

Syllabus Topic : Newton's Cotes Integration Formulas

4.2 Newton-Cotes Quadrature Formula

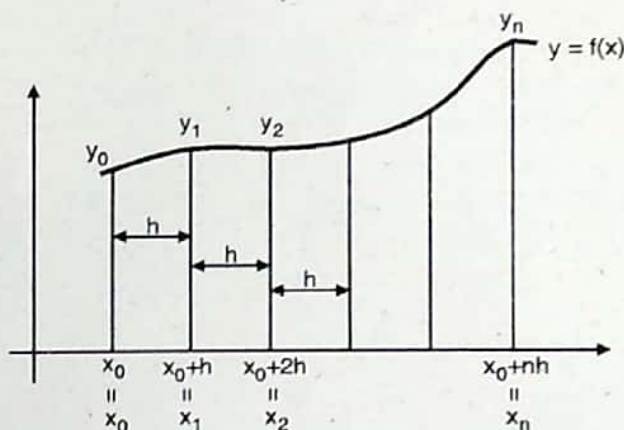


Fig. 4.2.1

- As shown in Fig. 4.2.1. The curve $y = f(x)$ is divided into n number of strips, with equal distance between each strip. The equal distance is called step size (h).

Let x_0 = Initial limit of the function $f(x)$ move here

and $x_1 = x_0 + h$

$x_2 = x_0 + 2h$

:

:

$x_n = x_0 + nh$ = final limit of the function $f(x)$

As $y = f(x)$

$\therefore y_0 = f(x_0)$

$\therefore y_1 = f(x_1)$

:

:

$y_n = f(x_n)$



- The quadrature formula for this curve becomes,

$$I = \int_{x_0}^{x_0+nh} f(x) \cdot dx = \int_{x_0}^{x_0+nh} f(x) dx$$

- Applying Newton's forward Interpolation then integrating we get,

$$I = \int_{x_0}^{x_0+nh} f(x) dx = nh \left[y_0 + \frac{n}{2} \Delta y_0 + \frac{n(2n-3)}{12} \Delta^2 y_0 + \frac{n(n-2)^2}{24} \Delta^3 y_0 + \left(\frac{n^4}{5} - \frac{3n^3}{2} + \frac{11n^2}{3} - 3n \right) \frac{\Delta^4 y_0}{24} + \left(\frac{n^5}{6} - 2n^4 + \frac{35n^3}{4} - \frac{50n^2}{3} + 12n \right) \frac{\Delta^5 y_0}{120} + \left(\frac{n^6}{7} - \frac{15n^5}{6} + 17n^4 - \frac{225n^3}{4} + \frac{274n^2}{3} - 60n \right) \frac{\Delta^6 y_0}{720} + \dots \right]$$

Where $(\Delta y_0 = y_1 - y_0, \Delta^2 y_0 = y_1 - 2y_0 + y_2,$

$\Delta^3 y_0 = y_0 - 3y_1 + 3y_2 + y_3)$

- The above equation is called Newton Cotes quadrature formula.

4.3 Various Numerical Methods for Solving Numerical Integration

- For Newton Cotes quadrature formula, we can substitute various value of n and we get different formulae which are used in this chapter, for solving integration problem.

When, $n = 1$ then the method is called Trapezoidal method.

$n = 2$ then the method is called Simpson's method

(1/3 Rule)

$n = 3$ then the method is called Simpson's method

(3/8 Rule)

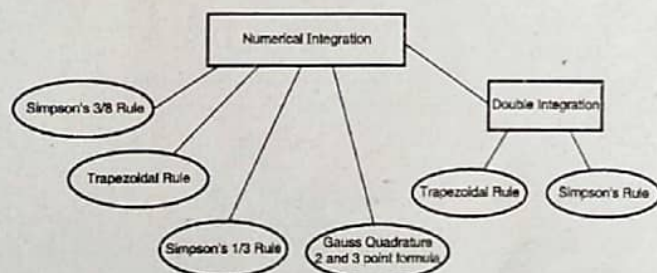


Fig. 4.3.1

Syllabus Topic : Trapezoidal Rule

4.4 Trapezoidal Rule

Review Question

Q. Explain the trapezoidal rule.

- If the curve is taking through the points (x_0, y_0) and (x_1, y_1) as a straight line i.e. A polynomial of first degree, then the Newton Cotes quadrature formula for $n = 1$ becomes,

$$A = \int_{x_0}^{x_0+h} f(x) dx = h \left(y_0 + \frac{1}{2} \Delta y_0 \right)$$

$$= h \left(y_0 + \frac{1}{2} (y_1 - y_0) \right)$$

$$A = \frac{h}{2} (y_0 + y_1)$$

...(4.4.1)

- As shown in Fig. 4.4.1 when the curve is considered as a straight line, the area under the straight line will be the approximation of an integral of the curve.

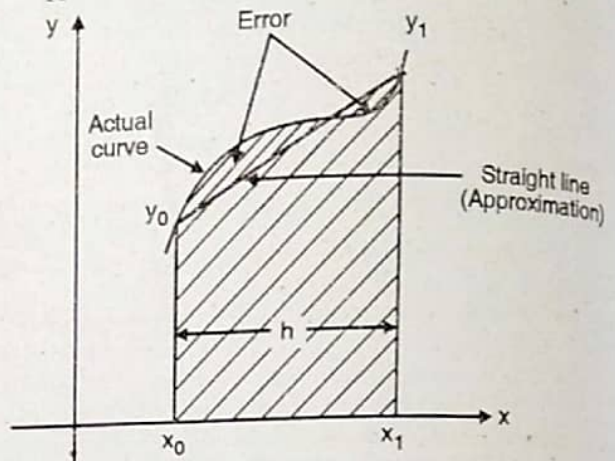


Fig. 4.4.1

- When we calculate integration of the curve, we have to consider the whole area under the curve, but according to the Trapezoidal Rule when we calculate area under straight line, As shown in Fig. 4.4.2 we calculate less area A_1 and additional area A_2 under a straight line. This will give slightly different answer than the actual area under the curve. Hence all these methods give approximate answer, inducing small error.

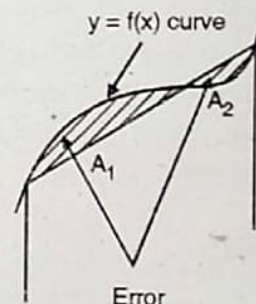


Fig. 4.4.2

- The local Truncation error is given by,

$$E = -\frac{1}{12} f''(\epsilon) \cdot h^3$$

Where ϵ lies somewhere in the interval x_0 and x_1 .

As shown in Fig. 4.4.3.

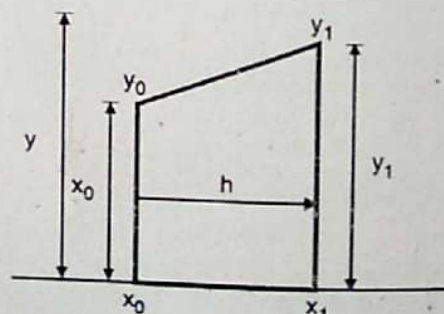


Fig. 4.4.3

- The area under a straight line is the area of Trapezoid and is given by,





$$\begin{aligned}\text{Area of Trapezoid} &= \frac{1}{2} \times (\text{Sum of parallel sides}) \times \text{distance} \\ &= \frac{1}{2} (y_0 + y_1) \times h \\ A &= \frac{h}{2} (y_0 + y_1)\end{aligned}$$

The above equation is Newton's cotes quadrature formula for $n = 1$ like Equation (4.4.1).

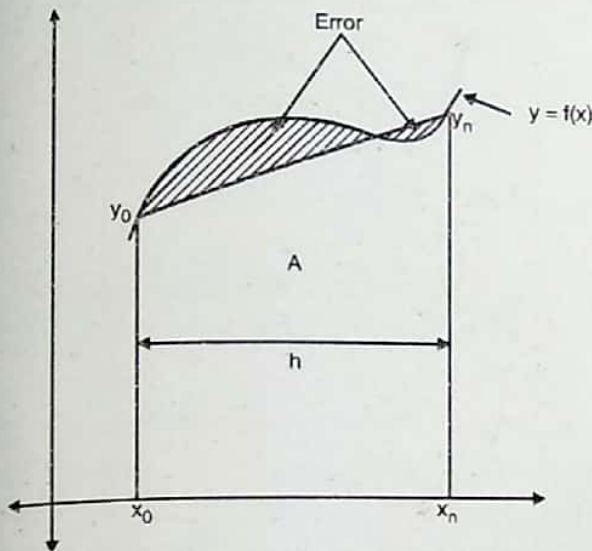


Fig. 4.4.4 : Curve is considered as a straight line with single strip

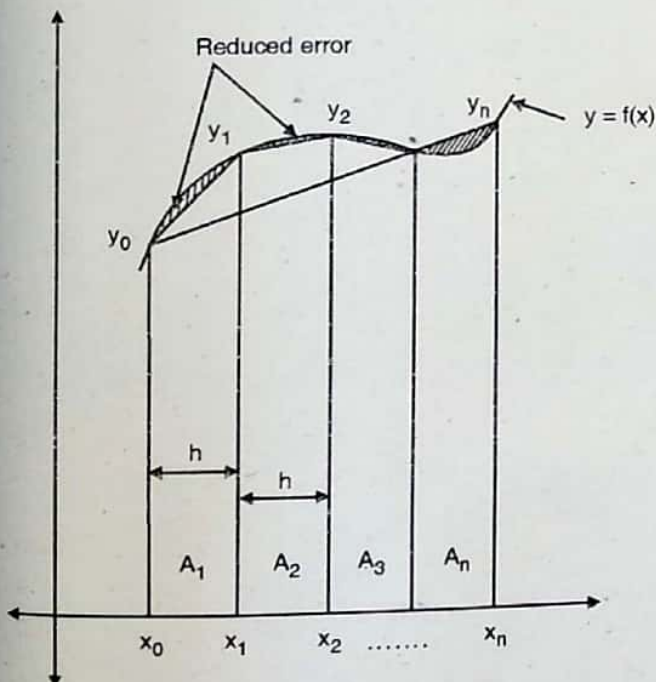


Fig. 4.4.5 : Curve is divided into n number of strips

From the Fig. 4.4.5 and Fig. 4.4.6 it is cleared that, when the curve is divided into n number of strips, then there is high reduction in the error.

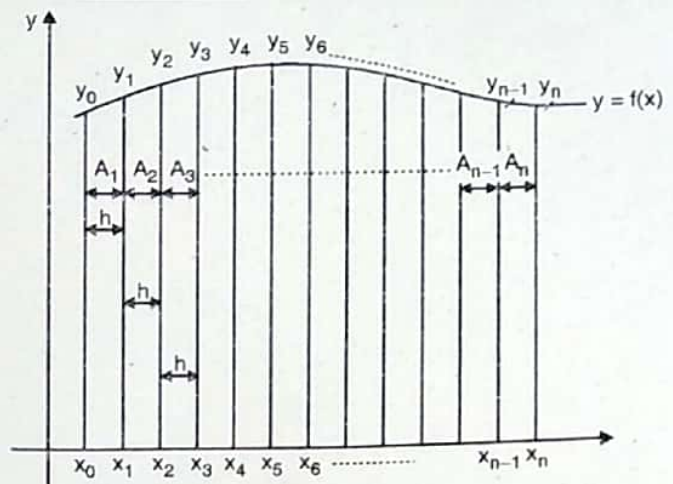


Fig. 4.4.6

From Fig. 4.4.6,

The total area under the curve is given by,

$$A = A_1 + A_2 + A_3 + \dots + A_n$$

Area A_1 can be calculated by applying Trapezoidal rule.

$$A_1 = \frac{h}{2} (y_0 + y_1)$$

$$\text{Similarly, } A_2 = \frac{h}{2} (y_1 + y_2)$$

$$A_3 = \frac{h}{2} (y_2 + y_3)$$

$$A_4 = \frac{h}{2} (y_3 + y_4)$$

$$A_4 = \frac{h}{2} (y_4 + y_5)$$

$$A_{n-1} = \frac{h}{2} (y_{n-2} + y_{n-1})$$

$$A_n = \frac{h}{2} (y_{n-1} + y_n)$$

$\therefore$ Total Area

$$\begin{aligned}A &= A_1 + A_2 + A_3 + \dots + A_{n-1} + A_n \\ &= \frac{h}{2} [(y_0 + y_1) + (y_1 + y_2) + (y_3 + y_4) + \dots \\ &\quad + (y_{n-2} + y_{n-1}) + (y_{n-1} + y_n)]\end{aligned}$$

If we observed the above equation we can say that,

- 1) Terms $y_1, y_2, y_3, \dots, y_{n-1}$ have repeated twice and
- 2) terms y_0 and y_n have repeated only once.

Hence a general formula for the area divided into n strips is given by,

$$A = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$$

$$A = \frac{h}{2} [(y_0 + y_n) + 2(\text{Remaining terms of } y)]$$

The above formula is called Trapezoidal formula to calculate area under the curve.



4.4.1 Stepwise Solution to Solve the Problems on Trapezoidal Rule (Illustrated by Numerical Example)

Example 4.4.1

Find the integration of $(4x + 2)$ in the limits 1 to 4 by Trapezoidal rule using six strips.

Solution :

Given data : Function $y = f(x) = 4x + 2$

Initial limit = $x_0 = 1$

Final limit = $x_n = 4$

Number of strips = $n = 6$

$$A = \int_{x_0}^{x_n} f(x) dx$$

$$A = \int_1^4 (4x + 2) \cdot dx$$

Step 1 : Calculate step size $\Rightarrow h :$

$$\therefore h = \frac{x_n - x_0}{n} = \frac{4 - 1}{6} = 0.5$$

Step 2 : Find the successive value of $x :$

$$\therefore x_0 = 1$$

$$x_1 = x_0 + h = 1 + 0.5 = 1.5$$

$$x_2 = x_0 + 2h = 1 + 1 = 2$$

$$x_3 = x_0 + 3h = 1 + 1.5 = 2.5$$

$$x_4 = x_0 + 4h = 1 + 2 = 3$$

$$x_5 = x_0 + 5h = 1 + 2.5 = 3.5$$

$$x_n = x_0 + nh = 1 + 3 = 4$$

Step 3 : Find the corresponding values of y for given value of $x :$

$$\therefore y = f(x) = 4x + 2$$

$$y_0 = f(x_0)$$

$$y_0 = (4(x_0) + 2) = 4(1) + 2$$

$$y_0 = 6$$

Similarly,

$$y_1 = f(x_1) = f(x_0 + h) \quad (\because x_1 = 1.5)$$

$$= 4(1.5) + 2 = 8$$

$$y_2 = f(x_2) = f(x_0 + 2h) \quad (\because x_2 = 2)$$

$$= 4(2) + 2 = 10$$

$$y_3 = f(x_3) = f(x_0 + 3h) \quad (\because x_3 = 2.5)$$

$$= 4(2.5) + 2 = 12$$

$$y_4 = f(x_4) = f(x_0 + 4h) \quad (\because x_4 = 3)$$

$$= 4(3) + 2 = 14$$

$$y_5 = f(x_5) = f(x_0 + 5h) \quad (\because x_5 = 3.5)$$

$$= 4(3.5) + 2 = 16$$

$$y_n = y_6 = f(x_6) = f(x_n)$$

$$= 4(4) + 2$$

$$y_n = 18$$

Step 4 : Apply the Trapezoidal rule to calculate the area.

$$A = A_1 + A_2 + A_3 + \dots + A_n$$

$$A = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$$

$$= \frac{0.5}{2} [(6 + 18) + 2(8 + 10 + 12 + 14 + 16)]$$

$$A = 36$$

$$\text{Area under } \int_1^4 f(x) = \int_1^4 4x + 2 = 36$$

...Ans.

Note : Flowchart and Program for Trapezoidal Rule are explained in Chapter 9.

Example 4.4.2

Find the integration of $\frac{1}{1+x^2}$ in the limits 0 to 1 by Trapezoidal rule using four strips.

Solution :

Given data : Function $y = f(x) = \frac{1}{1+x^2}$

Initial limit = $x_0 = 0$

Final limit = $x_n = 1$

Number of strips = $n = 4$

$$A = \int_{x_0}^{x_n} f(x) dx$$

$$A = \int_0^1 \frac{1}{1+x^2} \cdot dx$$

Step 1 : Calculate step size $\Rightarrow h :$

$$\therefore h = \frac{x_n - x_0}{n} = \frac{1 - 0}{4} = 0.25$$

Step 2 : Find the successive value of $x :$

$$\therefore x_0 = 0; \quad x_1 = x_0 + h = 0.25$$

$$x_2 = x_0 + 2h = 0.5; \quad x_3 = x_0 + 3h = 0.75$$

$$x_n = x_4 = x_0 + 4h = 1$$

Step 3 : Find the corresponding values of y for given value of x

$$\therefore y = f(x) = \frac{1}{1+x^2}$$

$$1) \quad y_0 = f(x_0) = f(0) \quad y_0 = \frac{1}{1+x_0^2} = 1$$

Similarly,

$$2) \quad y_1 = f(x_1) = f(0.25) = \frac{1}{1+x_1^2} = 0.94117$$



$$3) \quad y_2 = f(x_2) = f(0.5) = \frac{1}{1+x_2} = 0.8$$

$$4) \quad y_3 = f(x_3) = f(0.75) = \frac{1}{1+x_3} = 0.64$$

$$5) \quad y_4 = f(x_4) = f(1) = \frac{1}{1+x_4} = y_n = 0.5$$

Step 4: Apply Trapezoidal rule to calculate the area.

$$A = A_1 + A_2 + A_3 + \dots + A_n$$

$$A = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$$

$$= \frac{0.25}{2} [(1 + 0.5) + 2(0.9411 + 0.8 + 0.64)]$$

$$A = 0.7827$$

$$\text{Area under } f(x) \quad A = \int_0^1 \frac{1}{1+x^2} \cdot dx = 0.7827 \quad \dots \text{Ans.}$$

Example 4.4.3

Evaluate $\int_0^1 \sqrt{\sin(x) + \cos(x)} \cdot dx$ using Trapezoidal rule taking 8 divisions.

Solution:

Given data: Function $y = f(x) = \sqrt{\sin(x) + \cos(x)}$

Initial limit = $x_0 = 0$

Final limit = $x_n = 1$

Number of strips = $n = 8$

$$A = \int_{x_0}^{x_n} f(x) \, dx$$

$$A = \int_0^1 \sqrt{\sin(x) + \cos(x)} \cdot dx$$

Note: Convert the calculator on 'RAD' mode when the problem contains trigonometric function like (cos, sin).

Step 1: Calculate step size $\Rightarrow h$:

$$\therefore h = \frac{x_n - x_0}{n} = \frac{1-0}{8} = 0.125$$

Step 2: Find the successive value of x :

$$\therefore x_0 = 0 \quad ; \quad x_1 = x_0 + h = 0.125$$

$$x_2 = x_0 + 2h = 0.25 \quad ; \quad x_3 = x_0 + 3h = 0.375$$

$$x_4 = x_0 + 4h = 0.5 \quad ; \quad x_5 = x_0 + 5h = 0.625$$

$$x_6 = x_0 + 6h = 0.75 \quad ; \quad x_7 = x_0 + 7h = 0.875$$

$$x_n = x_8 = x_0 + 8h = 1$$

Step 3: Find the corresponding values of y for given value of x

$$\therefore y = f(x) = \sqrt{\sin(x) + \cos(x)}$$

$$1) \quad y_0 = f(x_0) = f(0)$$

$$y_0 = \sqrt{\sin(x_0) + \cos(x_0)}$$

$$y_0 = 1$$

Similarly,

$$2) \quad y_1 = f(x_1) = f(0.125) = \sqrt{\sin(x_1) + \cos(x_1)} = 1.0568$$

$$3) \quad y_2 = f(x_2) = f(0.25) = \sqrt{\sin(x_2) + \cos(x_2)} = 1.1028$$

$$4) \quad y_3 = f(x_3) = f(0.375) = \sqrt{\sin(x_3) + \cos(x_3)} = 1.1387$$

$$5) \quad y_4 = f(x_4) = f(0.5) = \sqrt{\sin(x_4) + \cos(x_4)} = 1.1649$$

$$6) \quad y_5 = f(x_5) = f(0.625) = \sqrt{\sin(x_5) + \cos(x_5)} = 1.1815$$

$$7) \quad y_6 = f(x_6) = f(0.75) = \sqrt{\sin(x_6) + \cos(x_6)} = 1.18888$$

$$8) \quad y_7 = f(x_7) = f(0.875) = \sqrt{\sin(x_7) + \cos(x_7)} = 1.1868$$

$$9) \quad y_8 = f(x_8) = f(1) = \sqrt{\sin(x_8) + \cos(x_8)} = y_n = 1.175$$

Step 4: Write the area for Trapezoidal rule:

$$A = A_1 + A_2 + A_3 + \dots + A_n$$

$$A = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$$

$$= \frac{0.125}{2} [(1 + 1.175) + 2(1.0568 + 1.10286$$

$$+ 1.13876 + 1.16490 + 1.18155 + 1.1888 + 1.18681)]$$

$$A = 1.13851$$

$$\text{Area under } f(x) \quad A = \int_0^1 \sqrt{\sin(x) + \cos(x)} \cdot dx = 1.138528 \dots \text{Ans.}$$

Example 4.4.4

Evaluate $\int_1^4 e^x + x^3 - 2x + 1 \, dx$ using Trapezoidal rule taking 6 divisions.

Solution:

Given data: Function $y = f(x) = e^x + x^3 - 2x + 1$

Initial limit = $x_0 = 1$

Final limit = $x_n = 4$

Number of strips = $n = 6$

$$A = \int_{x_0}^{x_n} f(x) \, dx$$

$$A = \int_1^4 e^x + x^3 - 2x + 1 \cdot dx$$

Step 1: Calculate step size $\Rightarrow h$:

$$\therefore h = \frac{x_n - x_0}{n} = \frac{4-1}{6} = 0.5$$

Step 2: Find the successive value of x :

$$\therefore x_0 = 1 \quad ; \quad x_1 = x_0 + h = 1.5$$

$$x_2 = x_0 + 2h = 2 \quad ; \quad x_3 = x_0 + 3h = 2.5$$

$$x_4 = x_0 + 4h = 3 \quad ; \quad x_5 = x_0 + 5h = 3.5$$



$$x_n = x_6 = x_0 + 6h = 4$$

Step 3: Find the corresponding values of y for given value of x

$$\therefore y = f(x) = e^x + x^3 - 2x + 1$$

$$(1) \quad y_0 = f(x_0) = f(1)$$

$$y_0 = e^1 + 1^3 - 2 \cdot 1 + 1$$

$$y_0 = 2.7182$$

Similarly,

$$(2) \quad y_1 = f(x_1) = f(1.5) = e^{1.5} + 1.5^3 - 2 \cdot 1.5 + 1 = 5.8566$$

$$(3) \quad y_2 = f(x_2) = f(2) = e^2 + 2^3 - 2 \cdot 2 + 1 = 12.389$$

$$(4) \quad y_3 = f(x_3) = f(2.5) = e^{2.5} + 2.5^3 - 2 \cdot 2.5 + 1 = 23.807$$

$$(5) \quad y_4 = f(x_4) = f(3) = e^3 + 3^3 - 2 \cdot 3 + 1 = 42.085$$

$$(6) \quad y_5 = f(x_5) = f(3.5) = e^{3.5} + 3.5^3 - 2 \cdot 3.5 + 1 = 69.99$$

$$(7) \quad y_6 = f(x_6) = f(4) = e^4 + 4^3 - 2 \cdot 4 + 1 = 111.598$$

Step 4: Write the area for Trapezoidal rule:

$$A = A_1 + A_2 + A_3 + \dots + A_n$$

$$A = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$$

$$= \frac{0.5}{2} [(2.7182 + 111.598) + 2(5.8566 + 12.389 + 23.807 + 42.085 + 69.99)]$$

$$A = 105.643$$

$$\text{Area under } f(x) A = \int_1^4 e^x - x^3 - 2x + 1 \, dx = 105.643 \quad \dots \text{Ans.}$$

Example 4.4.5

Find the integration of $\int_0^2 e^x \cdot \sin x \, dx$ by Trapezoidal rule using four strips.

Solution:

Given data: Function $y = f(x) = e^x \cdot \sin x$

$$\text{Initial limit} = x_0 = 0$$

$$\text{Final limit} = x_n = 2$$

$$\text{Number of strips} = n = 4$$

$$A = \int_{x_0}^{x_n} f(x) \, dx; \quad A = \int_0^2 e^x \cdot \sin x \, dx$$

Note: Convert the calculator on 'RAD' mode when the problem contains trigonometric function like (cos, sin).

Step 1: Calculate step size $\Rightarrow h$:

$$\therefore h = \frac{x_n - x_0}{n} = \frac{2 - 0}{4} = 0.5$$

Step 2: Find the successive value of x :

$$\therefore x_0 = 0, \quad x_1 = x_0 + h = 0.5, \quad x_2 = x_0 + 2h = 1$$

$$x_3 = x_0 + 3h = 1.5, \quad x_n = x_4 = x_0 + 4h = 2$$

Step 3: Find the corresponding values of y for given value of x

$$\therefore y = f(x) = e^x \cdot \sin x$$

$$(1) \quad y_0 = f(x_0) = f(0)$$

$$y_0 = e^0 \cdot \sin x_0 = 0$$

Similarly,

$$(2) \quad y_1 = f(x_1) = f(0.5) = e^{0.5} \cdot \sin x_1 = 0.790$$

$$(3) \quad y_2 = f(x_2) = f(1) = e^1 \cdot \sin x_2 = 2.287$$

$$(4) \quad y_3 = f(x_3) = f(1.5) = e^{1.5} \cdot \sin x_3 = 4.470$$

$$(5) \quad y_4 = f(x_4) = f(2) = e^2 \cdot \sin x_4 = y_n = 6.7188$$

Step 4: Write the area for Trapezoidal rule:

$$A = A_1 + A_2 + A_3 + \dots + A_n$$

$$A = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$$

$$= \frac{0.5}{2} [(0 + 6.7188) + 2(0.790 + 2.287 + 4.470)]$$

$$= 5.4538$$

$$\text{Area under } f(x) A = \int_0^2 e^x \cdot \sin x \, dx = 5.4538 \quad \dots \text{Ans.}$$

Example 4.4.6

Find the integration of $\int_{\pi/2}^{\pi} \frac{x \cos x}{\sin x} \, dx$ by Trapezoidal rule using four strips.

Solution:

Given data: Function $y = f(x) = \frac{x \cos x}{\sin x} \, dx$

$$\text{Initial limit} = x_0 = \pi/2 = 3.142/2 = 1.571$$

$$\text{Final limit} = x_n = \pi = 3.142$$

$$\text{Number of strips} = n = 4$$

$$A = \int_{x_0}^{x_n} f(x) \, dx$$

$$A = \int_{1.571}^{3.142} \frac{x \cos x}{\sin x} \, dx$$

Note: Convert the calculator on 'RAD' mode when the problem contains trigonometric function like (cos, sin).

Step 1: Calculate step size $\Rightarrow h$:

$$\therefore h = \frac{x_n - x_0}{n} = \frac{3.142 - 1.571}{4} = 0.393$$

Step 2: Find the successive value of x :

$$\therefore x_0 = 1.571; \quad x_1 = x_0 + h = 1.963$$

$$x_2 = x_0 + 2h = 2.356; \quad x_3 = x_0 + 3h = 2.749$$

$$x_n = x_4 = x_0 + 4h = 3.142.$$



Step 3: Find the corresponding values of y for given value of x

$$y = f(x) = \frac{x \cos x}{\sin x}$$

$$(1) \quad y_0 = f(x_0) = f(1.571) = \frac{x_0 \cos x_0}{\sin x_0} = 0.00125 \quad \text{and}$$

Similarly,

$$(2) \quad y_1 = f(x_1) = f(1.9631) = \frac{x_1 \cos x_1}{\sin x_1} = -0.81196$$

$$(3) \quad y_2 = f(x_2) = f(2.356) = \frac{x_2 \cos x_2}{\sin x_2} = -2.355$$

$$(4) \quad y_3 = f(x_3) = f(2.749) = \frac{x_3 \cos x_3}{\sin x_3} = -6.638$$

$$(5) \quad y_4 = f(x_4) = f(3.142) = \frac{x_4 \cos x_4}{\sin x_4} = y_n = 7714.09$$

Step 4: Write the area for Trapezoidal rule :

$$A = A_1 + A_2 + A_3 + \dots + A_n$$

$$A = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$$

$$= \frac{0.393}{2} [(0.00125 + 7714.09)$$

$$+ 2(-0.8119 + (-2.355) + (-6.638))]$$

$$A = 1511.96$$

$$\text{Area under } f(x) \quad A = \int_{\pi/2}^{\pi} \frac{x \cos x}{\sin x} dx = 1511.96 \quad \dots \text{Ans.}$$

Example 4.4.7

Evaluate $\int_0^2 \frac{x}{\sqrt{2+x^2}} dx$ by using Trapezoidal rule with four strips.

Solution :

Given data :

$$\text{Function } i = f(x) = \frac{x}{\sqrt{2+x^2}}$$

$$\text{Initial limit} = 0 \quad \therefore \text{Final limit} = 2$$

$$\text{Number of strips} = n = 4$$

$$n = \int_0^2 \frac{x}{\sqrt{2+x^2}} dx$$

Step 1: Calculate step size, h

$$\therefore h = \frac{x_n - x_0}{n} = \frac{2-0}{4} = 0.5$$

Step 2: Find the successive value of x .

$$x_0 = 0,$$

$$x_1 = x_0 + h = 0 + 0.5 = 0.5$$

$$x_2 = x_0 + 2h = 0 + 2(0.5) = 1$$

$$x_3 = x_0 + 3h = 0 + 3(0.5) = 1.5$$

$$x_n = x_4 = x_0 + 4h = 0 + 4(0.5) = 2$$

Step 3: Find the corresponding value of y for given value of x

$$y = F(x) = \frac{x}{\sqrt{2+x^2}}$$

$$(1) \quad y_0 = F(x_0) = F(0) = \frac{0}{\sqrt{2+0^2}} = 0$$

Similarly

$$(2) \quad y_1 = F(x_1) = F(0.5) = \frac{x}{\sqrt{2+x^2}} = \frac{0.5}{\sqrt{2+(0.5)^2}} = 0.333$$

$$(3) \quad y_2 = F(x_2) = F(1) = \frac{x}{\sqrt{2+x^2}} = \frac{1}{\sqrt{2+(1)^2}} = 0.577$$

$$(4) \quad y_3 = F(x_3) = F(1.5) = \frac{x}{\sqrt{2+x^2}} = \frac{1.5}{\sqrt{2+(1.5)^2}} = 0.727$$

$$(5) \quad y_4 = F(x_4) = F(2) = \frac{x}{\sqrt{2+x^2}} = \frac{2}{\sqrt{2+2^2}} = 0.816$$

Step 4: Apply trapezoidal rule to calculate the area

$$A = A_1 + A_2 + A_3 + \dots + A_n$$

$$A = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$$

$$= \frac{0.5}{2} [(0 + 0.816) + 2(0.333 + 0.577 + 0.727)]$$

$$A = 3.4807$$

$$\text{Area under } P(x) \quad A = \int_0^2 \frac{x}{\sqrt{2+x^2}} dx = 1.022$$

Example 4.4.8

Evaluate $\int_2^6 \log_{10}(x) \cdot dx$ by using Trapezoidal rule with eight strips.

Solution :

Given data :

$$\text{Function } y = f(x) = \log_{10} x$$

$$\text{Initial limit} = x_0 = 2 \quad \therefore \text{Final limit} = x_n = 6$$

$$\text{Number of strip} = 8$$

$$A = \int_2^6 \log_{10} x dx$$

Step 1: Calculate step size $\Rightarrow h$:

$$\therefore h = \frac{x_n - x_0}{n} = \frac{6-2}{8} = 0.5$$

Step 2: Find the successive value of x ,

$$x_0 = 2 \quad ; \quad x_1 = x_0 + h = 2 + 0.5 = 2.5$$

$$x_2 = x_0 + 2h = 3 \quad ; \quad x_3 = x_0 + 3h = 3.5$$

$$x_4 = x_0 + 4h = 4 \quad ; \quad x_5 = x_0 + 5h = 4.5$$

$$x_6 = x_0 + 6h = 5 \quad ; \quad x_7 = x_0 + 7h = 5.5$$

$$x_8 = x_n = x_0 + 8h = 6$$

Step 3: Find the corresponding value of y for given value of x

$$y = f(x) = \log_{10} x$$

$$y_0 = f(x_0) = f(2) = 0.30102$$

$$y_1 = f(x_1) = f(2.5) = 0.3979$$

$$y_2 = f(x_2) = f(3) = 0.4771$$

$$y_3 = f(x_3) = f(3.5) = 0.5440$$

$$y_4 = f(x_4) = f(4) = 0.6020$$